

PARTIAL DIFFERENTIAL EQUATIONS (PDES)

Outline

- Introduction
- Finite-difference methods

What is a PDE?

- PDE is a differential equation with **more than 1 independent variable**
- PDEs are more difficult to solve than ODEs due to their added complexity
- PDEs generally involve multiple spatial dimensions and time
- Many problems in civil engineering involve PDEs
 - Groundwater flow and transport, structural analysis, air quality, surface water flow

Classification of PDEs

- Elliptic PDEs
 - Steady State Groundwater Flow Equation
 - Electromagnetic field equation
 - Other forms of Laplace equation
- Parabolic PDEs
 - Unsteady state groundwater flow equation
 - Convection-dispersion equation
 - Heat equation
- Hyperbolic PDEs
 - Convection equation
 - Wave equation

Classification of PDEs (contd).

- General 2nd order homogeneous PDE with 2 independent variables (x, y) is of the form

$$Af_{xx} + Bf_{xy} + Cf_{yy} + Df_x + Ef_y + Ff = G$$

- Classification is according to the value of $B^2 - 4AC$
 - Elliptic: $B^2 - 4AC < 0$
 - Parabolic: $B^2 - 4AC = 0$
 - Hyperbolic: $B^2 - 4AC > 0$
- Classification of higher order PDEs and those with more than 2 independent variables is more complex

Classification of PDEs (contd).

- For 2D Laplace equation

$$\nabla^2 u - f = 0$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} - f(x, y) = 0$$

- $A = 1, B = 0, C = 1, B^2 - 4AC = -1 < 0$. Therefore, 2D Laplace equation is **elliptic**.
- Note that we take 2D case to restrict ourselves to 2 independent variables.

Classification of PDEs (contd).

- For 1D Wave equation

$$\frac{\partial^2 u}{\partial t^2} - \nabla^2 u - f = 0$$

$$\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} - f = 0$$

- $A = 1, B = 0, C = -1, B^2 - 4AC = 1 > 0$. Therefore, 1-D Wave equation is **hyperbolic**.
- We take 1D case to restrict ourselves to 2 independent variables.

Classification of PDEs (contd).

- For 1D Diffusion equation

$$\frac{\partial u}{\partial t} - \nabla^2 u - f = 0$$

$$\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} - f = 0$$

- $A = 1, B = C = 0, B^2 - 4AC = 0$. Therefore, 1-D Diffusion equation is **parabolic**.
- We take 1D case to restrict ourselves to 2 independent variables.

Classification of PDEs (contd.)

- Generally time-independent or steady state problems such as steady state groundwater flow, electromagnetic field equation are elliptic and numerous other problems defined by Poisson equation or Laplace equation are **elliptic**.
- Time-dependent or unsteady problems are generally **parabolic** or **hyperbolic**.

Classification of PDEs

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TABLE PT8.1 Categories into which linear, second-order partial differential equations in two variables can be classified.

$B^2 - 4AC$	Category	Example
< 0	Elliptic	Laplace equation (steady state with two spatial dimensions) $\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$
$= 0$	Parabolic	Heat conduction equation (time variable with one spatial dimension) $\frac{\partial T}{\partial t} = k' \frac{\partial^2 T}{\partial x^2}$
> 0	Hyperbolic	Wave equation (time variable with one spatial dimension) $\frac{\partial^2 y}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 y}{\partial t^2}$

Domain of Dependence and Domain of Influence

- *Domain of dependence* of any point P is defined as the region of the solution domain upon which the solution at point P , $f(x_p, y_p) = f_p$ depends. In other words, f_p depends on everything that has happened in the domain of dependence.
- *Range of influence* of any point P is defined as the region of the solution domain in which the solution $f(x, y)$ is influenced by the solution at P . In other words f_p influences the solution at all points in the range of influence.

Elliptic PDES

- Steady State Groundwater Flow Equation
 - Uniform hydraulic conductivity (permeability) case

$$K\nabla^2 h = q$$

$$k_{xx} \frac{\partial^2 h}{\partial x^2} + k_{yy} \frac{\partial^2 h}{\partial y^2} = q(x, y)$$

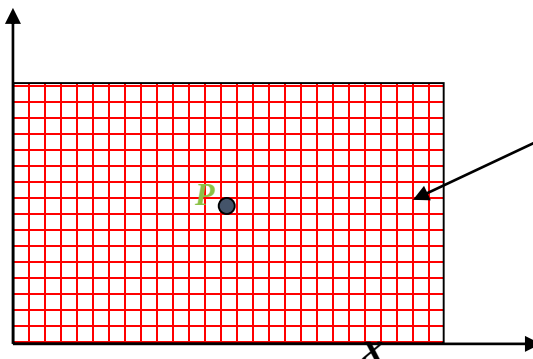
- Non-uniform hydraulic conductivity case

$$\nabla \cdot (K \nabla h) = q$$

$$\frac{\partial}{\partial x} \left(k_{xx} \frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_{yy} \frac{\partial h}{\partial y} \right) = q(x, y)$$

Elliptic PDEs

- Signal propagation speed is **infinite**. Any perturbation at point P is felt immediately throughout the domain.
- Domain of dependence is entire domain.
 - solution at any point P depends on solution at all other points.
- Domain of influence is entire domain.
 - solution at any point P influences solution at all other points.



*Range of Influence
Domain of Dependence
Entire Domain*

Parabolic PDES

- Unsteady State Groundwater Flow Equation
 - Uniform hydraulic conductivity (permeability) case

$$S_s \frac{\partial h}{\partial t} = K \nabla^2 h - q$$

$$S_s \frac{\partial h}{\partial t} = k_x \frac{\partial^2 h}{\partial x^2} + k_y \frac{\partial^2 h}{\partial y^2} - q(x, y)$$

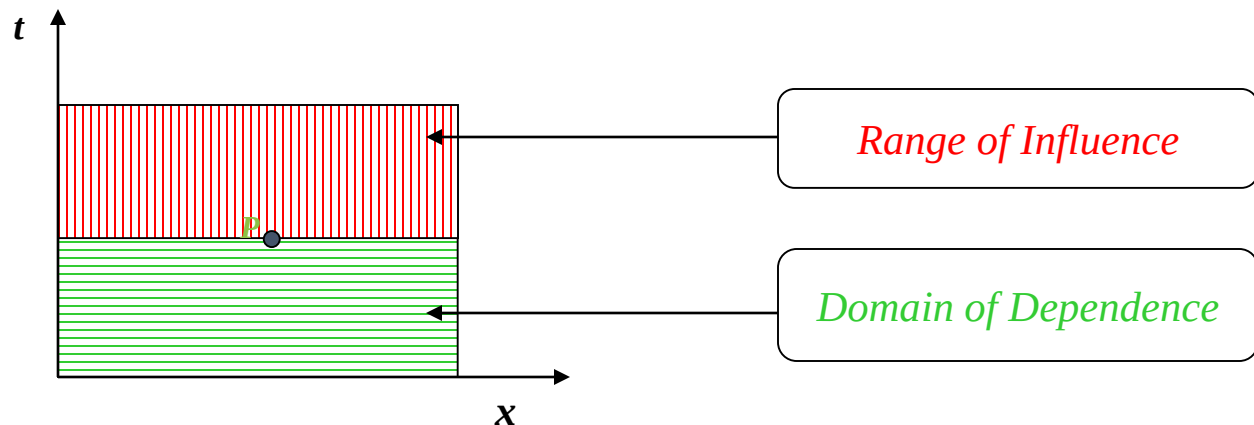
- Non-uniform hydraulic conductivity case

$$S_s \frac{\partial h}{\partial t} = \nabla \cdot (K \nabla h) - q$$

$$S_s \frac{\partial h}{\partial t} = \frac{\partial}{\partial x} \left(k_x \frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_y \frac{\partial h}{\partial y} \right) - q(x, y)$$

Parabolic PDEs

- Signal propagation speed is **finite**. Any perturbation at point P travels at a finite speed $u = \Delta x / \Delta t$ to all other points.
- Domain of dependence is entire spatial domain in the past time domain.
 - solution at any point P depends on solution at all points in the previous time steps.
- Domain of influence is entire spatial domain in the future time domain.
 - solution at any point P influences solution at all points in the future time steps.



Hyperbolic PDEs

- Convection equation

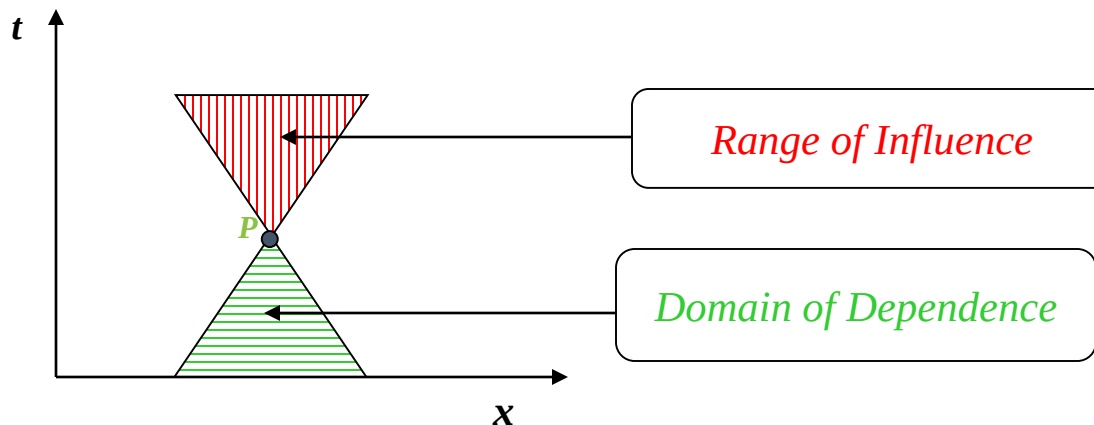
$$\frac{\partial c}{\partial t} = -\mathbf{v} \cdot \nabla c$$

- Wave equation

$$\frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} = \nabla \cdot (\rho \nabla p)$$

Hyperbolic PDEs

- Signal propagation speed is **finite**. Any perturbation at point **P** travels at a finite speed $u = \Delta x / \Delta t$ to all other points.
- Domain of dependence is only part of the spatial domain in the past time domain.
 - solution at any point **P** depends on solution at selected points in the previous time steps.
- Domain of influence is only part of the spatial domain in the future time domain.
 - solution at any point **P** influences solution at selected points in the future time steps.



Convection-Diffusion Equation

- CDE is classified as a parabolic PDE but can exhibit parabolic or hyperbolic behavior depending on PDE coefficients

$$\frac{\partial c}{\partial t} = \mathbf{D} \nabla^2 c - \mathbf{v} \nabla c \quad \text{Homogeneous}$$

$$\frac{\partial c}{\partial t} = \nabla (\mathbf{D} \nabla c) - \nabla (\mathbf{v} c) \quad \text{Inhomogeneous}$$

- $\mathbf{D}$ is the dispersion tensor, $c(x,y,z,t)$ is the solute concentration, and $\mathbf{v}(x,y,z,t)$ is the fluid velocity vector
- If $\mathbf{D}$ is large or $\mathbf{v}$ is small then it exhibits parabolic behavior
- If $\mathbf{D}$ is small or $\mathbf{v}$ is large then it exhibits hyperbolic behavior
- Different solution techniques may need to be adopted for each case

Convection Diffusion Equation

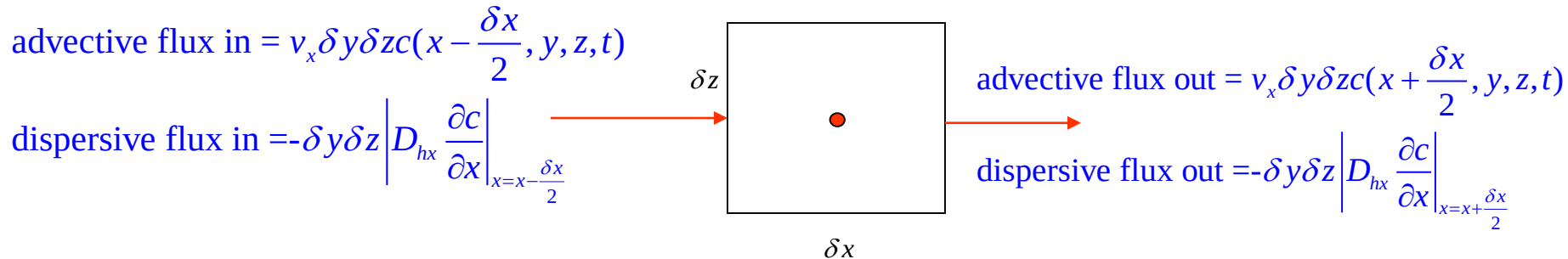
- CDE occurs in many areas of environmental engineering
 - groundwater contaminant transport, chemical transport in reactors, air pollution transport, chemical transport in rivers and estuaries
- Solutes can be transported by two mechanisms
 - Dispersion (includes diffusion and hydrodynamic dispersion)
 - Convection
- In addition solutes can undergo reactions
 - decay, sorption, bioremediation, reversible ion exchange
 - when reactions are added, the CDE is known as CDR convection-diffusion-reaction equation or ADR advection-diffusion-reaction equation

CDE (contd.)

- CDE for solute transport can be derived by simple mass balance for concentration in a fluid element
- time rate of change of solute mass in the fluid element = net rate of **solute influx** into the fluid element due to **dispersion** and **advection**
- Additional terms can be added to CDE for reactions and source sink terms

$$\frac{\partial c}{\partial t} = \mathbf{D} \nabla^2 c - \mathbf{v} \nabla c + R + qc^*$$

CDE Derivation



$$\underbrace{\frac{\partial(c \delta x \delta y \delta z)}{\partial t}}_{\frac{\partial m_{cv}}{\partial t}} = \underbrace{\left(v_x \delta y \delta z c(x - \frac{\delta x}{2}, y, z, t) - \delta y \delta z \left[D_{hx} \frac{\partial c}{\partial x} \right]_{x=x-\frac{\delta x}{2}} \right)}_{\dot{m}_{in}} - \underbrace{\left(v_x \delta y \delta z c(x + \frac{\delta x}{2}, y, z, t) - \delta y \delta z \left[D_{hx} \frac{\partial c}{\partial x} \right]_{x=x+\frac{\delta x}{2}} \right)}_{\dot{m}_{out}}$$

$$\frac{\partial c}{\partial t} = \nabla(\mathbf{D} \nabla c) - \nabla(\mathbf{v} c)$$

1D Convection Diffusion Equation

- Homogeneous CDE

$$\frac{\partial c}{\partial t} = \mathbf{D} \nabla^2 c - \mathbf{v} \nabla c$$

- Homogeneous 1-dimensional CDE is the simplest form

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} - v \frac{\partial c}{\partial x}$$

- For groundwater solute transport dispersion coefficient D is given by

$$D = D_m + \alpha v$$

D_m = molecular diffusivity
(L²/T)

α = dispersivity of medium (L)

Analytical Solutions to 1D CDE

- Governing Equation and initial and boundary conditions – 1D column

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} - v \frac{\partial c}{\partial x}$$

$$c(x, 0) = 0, \quad x > 0$$

$$c(0, t) = C_0, \quad t \geq 0$$

$$c(\infty, t) = 0, \quad t \geq 0$$

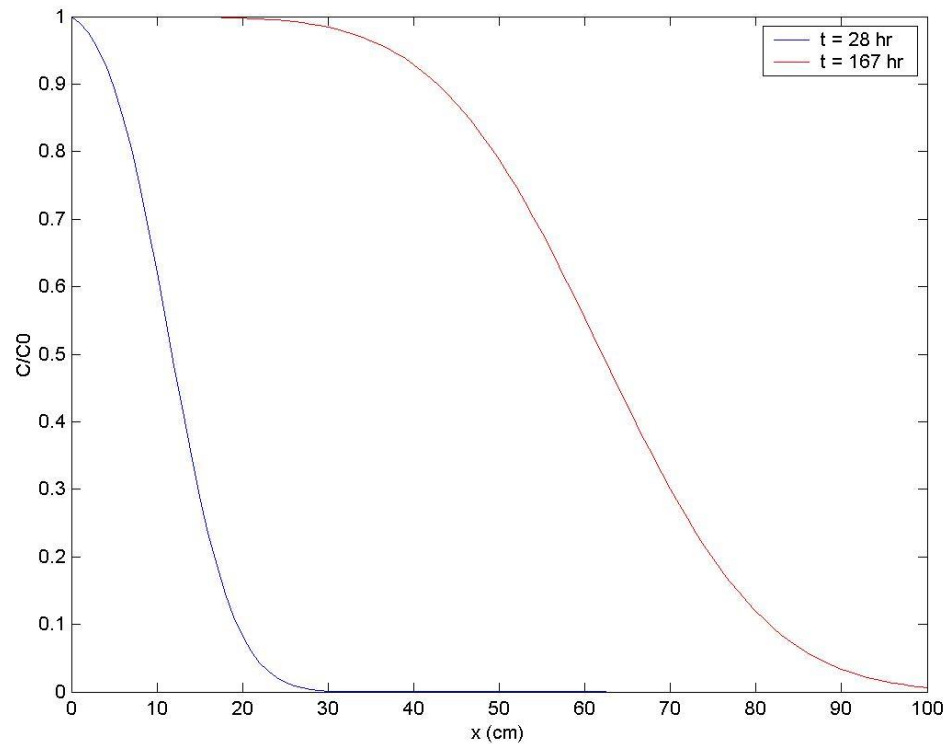


$$\text{erfc}(x) = 1 - \frac{1}{\sqrt{2\pi}} \int_0^x e^{-t^2} dt$$

- Analytical Solution

$$\frac{c(x, t)}{C_0} = \frac{1}{2} \left[\text{erfc} \left(\frac{x - vt}{2\sqrt{Dt}} \right) + \exp \left(\frac{vx}{D} \right) \text{erfc} \left(\frac{x + vt}{2\sqrt{Dt}} \right) \right]$$

Analytical Solution



$$V = 0.36 \text{ cm/hr}$$
$$\alpha = 2 \text{ cm}$$

FINITE DIFFERENCE METHODS FOR PDES

Finite-Difference Methods for PDEs

- A finite difference approximation (FDA) to any PDE can be obtained by applying various difference techniques to each term in the PDE.
 - The resulting equation is called the finite difference equation (FDE).
- A mix of Forward, Backward, Central (or higher order variants) differencing can be applied to the temporal and spatial derivatives in a PDE.
- For finite difference methods to be effective, the following criteria need to be met
 - Consistency
 - Stability
 - Convergence

Consistency, Order, Stability, and Convergence

- Consistency
 - An FDE is consistent with a PDE if the difference between the FDE and PDE (i.e., the truncation error) vanishes as the size of the grid spacing goes to zero independently.
- Order
 - The order of a FDA of a PDE is the rate at which the global error of the finite difference solution approaches zero as the size of the grid spacing approaches zero.
- Stability
 - When applied to a PDE that has a bounded solution, an FDE is stable if it produces a bounded solution and is unstable if it produces an unbounded solution.
- Convergence
 - A finite difference method is convergent if the solution of the FDE approaches the exact solution of the PDE as the size of the grid spacing goes to zero.

Consistency, Order, Stability, and Convergence (contd.)

- Lax Equivalence Theorem
 - Given a properly posed linear initial value problem and a FDA to it that is consistent, stability is the necessary and sufficient condition for convergence.
- Consistency can be checked by developing the modified difference equation (MDE)
 - MDE can be developed by expressing each term in the FDE in a Taylor series.
- Stability can be investigated using the Von-Neumann method (p. 491, Hoffman).

Implicit and Explicit Methods

- If the unknown of a time-dependent FDE appears on both the space and time derivatives then the method is implicit. If the unknown appears only on the time derivative then the method is explicit.
- Implicit methods require matrix solution. Explicit methods do not require matrix solution.
- Although not strictly defined, FDE of time-independent problems are usually implicit. However, the method of solution of the FDE (e.g., Jacobi method) may make the process explicit.

Implicit and Explicit Methods (contd.)

- Elliptic PDEs
 - Because the physical signal propagation speed is infinite for elliptic PDEs, implicit type methods (those that involve matrix solution) which propagate the solution at infinite speed are better suited for elliptic PDEs.
- Parabolic PDEs
 - Both implicit and explicit methods can be used for parabolic PDEs. Implicit methods are more common.
- Hyperbolic PDEs
 - Explicit methods are more popular for hyperbolic PDEs since the physical signal propagation speed is finite, and explicit methods also propagate the solution at finite speed.

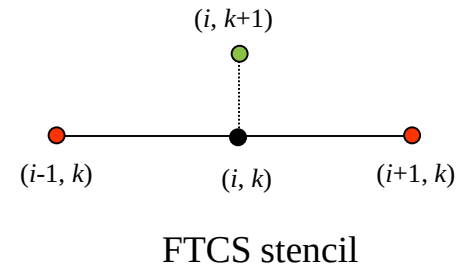
Finite Difference Approximation of CDE

- forward-time centered-space (FTCS)
- ODE Approach - Runge-kutta (ode45)
- backward-time centered-space (BTCS)
- Crank Nicolson
- Upwind Finite Difference Scheme

Forward-Time Centered-Space Scheme

- 1D CDE

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} - v \frac{\partial c}{\partial x}$$



- Discretize temporally and spatially
- For $i = 1, \dots, n$ and $k = 1, \dots, nt$

$$\frac{c_i^{k+1} - c_i^k}{\Delta t} = D \left(\frac{c_{i+1}^k - 2c_i^k + c_{i-1}^k}{\Delta x^2} \right) - v \left(\frac{c_{i+1}^k - c_{i-1}^k}{2\Delta x} \right)$$

$$c_i^{k+1} = c_i^k + \frac{D\Delta t}{\Delta x^2} (c_{i+1}^k - 2c_i^k + c_{i-1}^k) - \frac{v\Delta t}{2\Delta x} (c_{i+1}^k - c_{i-1}^k)$$

Stability and Accuracy of FTCS Scheme

- FDE for FTCS can be rewritten as

$$\begin{aligned}c_i^{k+1} &= c_i^k + DN(c_{i+1}^k - 2c_i^k + c_{i-1}^k) - \frac{CN}{2}(c_{i+1}^k - c_{i-1}^k) \\&= \left(DN + \frac{CN}{2}\right)c_{i-1}^k + (1 - 2DN)c_i^k + \left(DN - \frac{CN}{2}\right)c_{i+1}^k\end{aligned}$$

where

$$CN = \frac{v\Delta t}{\Delta x} = \text{convection number}$$

$$DN = \frac{D\Delta t}{\Delta x^2} = \text{diffusion number}$$

- FTCS method is conditionally stable. For stability it can be proven that $CN^2 \leq 2DN \leq 1$

Stability and Accuracy of FTCS Scheme (contd)

- From the previous expression we can arrive at the following famous condition (strictly valid for pure convection equation)

$$\text{courant or CFL condition: } \left(\frac{v\Delta t}{\Delta x} \right) \leq 1$$

- For accuracy (overshoot problem), we can arrive at the following condition

$$P = \text{grid peclet number} = \left(\frac{v\Delta x}{D} \right) \leq 2$$

ODE approach

- 1D CDE

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} - v \frac{\partial c}{\partial x}$$

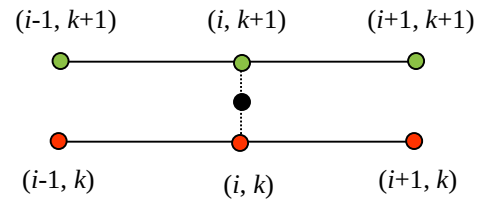
- Discretize spatial derivatives and solve system of ODEs

$$\frac{\partial c_i}{\partial t} = D \left(\frac{c_{i+1} - 2c_i + c_{i-1}}{\Delta x^2} \right) - v \left(\frac{c_{i+1} - c_{i-1}}{2\Delta x} \right) \quad i = 1, \dots, n$$

Crank-Nicolson Scheme

- Crank-Nicolson scheme approximates all derivatives at time $k+1/2$

$$\frac{c_i^{k+1} - c_i^k}{\Delta t} = \frac{D}{2} \left(\frac{c_{i+1}^{k+1} - 2c_i^{k+1} + c_{i-1}^{k+1}}{\Delta x^2} + \frac{c_{i+1}^k - 2c_i^k + c_{i-1}^k}{\Delta x^2} \right) - \frac{v}{2} \left(\frac{c_{i+1}^{k+1} - c_{i-1}^{k+1}}{2\Delta x} + \frac{c_{i+1}^k - c_{i-1}^k}{2\Delta x} \right)$$



Crank-Nicolson stencil

- C-N scheme is an implicit scheme. A linear system of equations of the form $Ax=b$ needs to be solved.

Stability and Accuracy of Crank-Nicolson

- Crank-Nicolson scheme is unconditionally stable and convergent.
- However, for accuracy the grid Peclet number condition generally needs to be met.

$$P = \text{grid peclet number} = \left(\frac{v\Delta x}{D} \right) \leq 2$$

Implementation of FTCS, BTCS, CN

- A common form of the FD equation is usually employed to implement the above schemes in 1 code

$$\frac{c_i^{k+1} - c_i^k}{\Delta t} = D \left(\omega \frac{c_{i+1}^{k+1} - 2c_i^{k+1} + c_{i-1}^{k+1}}{\Delta x^2} + (1 - \omega) \frac{c_{i+1}^k - 2c_i^k + c_{i-1}^k}{\Delta x^2} \right) - v \left(\omega \frac{c_{i+1}^{k+1} - c_{i-1}^{k+1}}{2\Delta x} + (1 - \omega) \frac{c_{i+1}^k - c_{i-1}^k}{2\Delta x} \right)$$

$$\omega = 0, \quad \text{FTCS}$$

$$\omega = 1, \quad \text{BTCS}$$

$$\omega = 0.5, \quad \text{CN}$$

Implementation of FTCS, BTCS, CN (contd.)

- For any node i , FDA can be rewritten as,

$$\begin{aligned}
 & -\omega \left(DN + \frac{CN}{2} \right) c_{i-1}^{k+1} + [1 + 2DN\omega] c_i^{k+1} - \omega \left(DN - \frac{CN}{2} \right) c_{i+1}^{k+1} = \\
 & (1-\omega) \left(DN + \frac{CN}{2} \right) c_{i-1}^k + [1 - 2DN(1-\omega)] c_i^k + (1-\omega) \left(DN - \frac{CN}{2} \right) c_{i+1}^k \\
 & d_i c_{i-1}^{k+1} + e_i c_i^{k+1} + f_i c_{i+1}^{k+1} = b_i
 \end{aligned}$$

$$\begin{pmatrix} \ddots & \ddots & & & 0 \\ \ddots & \ddots & \ddots & & \\ & d_i & e_i & f_i & \\ 0 & & \ddots & \ddots & \ddots \end{pmatrix}$$

- Where

$$d_i = -\omega \left(DN + \frac{CN}{2} \right), \quad e_i = 1 + 2DN\omega, \quad f_i = -\omega \left(DN - \frac{CN}{2} \right)$$

$$b_i = (1-\omega) \left(DN + \frac{CN}{2} \right) c_{i-1}^k + [1 - 2DN(1-\omega)] c_i^k + (1-\omega) \left(DN - \frac{CN}{2} \right) c_{i+1}^k, \quad i = 1, \dots, n$$

- The above values need to be modified for boundary nodes based on the appropriate boundary conditions.

First order upwind FD Scheme

- The basic concept behind upwind methods is that the convection derivative is approximated by one-sided FD approximation in the upwind direction (i.e., the direction from which the velocity comes).
- The simplest upwind FD scheme is a slight modification of the FTCS scheme. If v is in the positive x -direction ($v > 0$),

$$\frac{c_i^{k+1} - c_i^k}{\Delta t} = D \left(\frac{c_{i+1}^k - 2c_i^k + c_{i-1}^k}{\Delta x^2} \right) - v \left(\frac{c_i^k - c_{i-1}^k}{\Delta x} \right)$$
$$c_i^{k+1} = c_i^k + \frac{D\Delta t}{\Delta x^2} (c_{i+1}^k - 2c_i^k + c_{i-1}^k) - \frac{v\Delta t}{\Delta x} (c_i^k - c_{i-1}^k)$$

Upwind Schemes

- General form of first order upwind scheme (valid for any v)

$$\frac{c_i^{k+1} - c_i^k}{\Delta t} = D \left(\frac{c_{i+1}^k - 2c_i^k + c_{i-1}^k}{\Delta x^2} \right) - v \left(\frac{(1-\alpha)c_{i+1}^k + \beta c_i^k - \alpha c_{i-1}^k}{\Delta x} \right)$$

$$\beta = \text{sign}(v), \quad \alpha = \frac{1 + \beta}{2}$$

- When v is large upwind schemes provide reasonably stable solutions at the expense of accuracy (implicit numerical diffusion).
- Implicit and higher order upwind schemes can also be derived.

Initial and Boundary Conditions

- Initial condition

$$c(x, y, z, 0) = f(x, y, z)$$

Where $f(x, y, z)$ is a known function. Usually a constant concentration is assumed throughout the domain of interest

- Boundary conditions
 - Dirichlet or prescribed concentration (1st type)
 - Neuman or prescribed flux (2nd type)
 - Cauchy or mixed b.c (3rd type)

Dirichlet Boundary Condition (1st type)

- Prescribe a concentration along a portion of the boundary

$$c = c_0(x, y, z, t)$$

- Where $c_0(x, y, z, t)$ is a known function of space and time over that portion of boundary

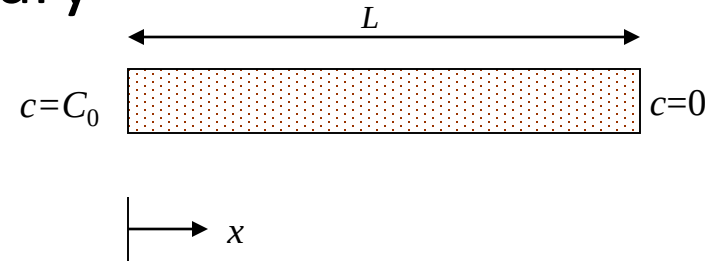
- Example

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} - v \frac{\partial c}{\partial x}$$

$$c(x, 0) = 0, \quad x > 0$$

$$c(0, t) = C_0, \quad t \geq 0$$

$$c(L, t) = 0, \quad t \geq 0$$



Example Problem with Dirichlet conditions

- General FDE given by

$$\begin{aligned}
 \frac{c_i^{k+1} - c_i^k}{\Delta t} = & D \left(\omega \frac{c_{i+1}^{k+1} - 2c_i^{k+1} + c_{i-1}^{k+1}}{\Delta x^2} + (1-\omega) \frac{c_{i+1}^k - 2c_i^k + c_{i-1}^k}{\Delta x^2} \right) \\
 & - v \left(\omega \frac{c_{i+1}^{k+1} - c_{i-1}^{k+1}}{2\Delta x} + (1-\omega) \frac{c_{i+1}^k - c_{i-1}^k}{2\Delta x} \right) \\
 & - \omega \left(DN + \frac{CN}{2} \right) c_{i-1}^{k+1} + [1 + 2DN\omega] c_i^{k+1} - \omega \left(DN - \frac{CN}{2} \right) c_{i+1}^{k+1} = \\
 & (1-\omega) \left(DN + \frac{CN}{2} \right) c_{i-1}^k + [1 - 2DN(1-\omega)] c_i^k + (1-\omega) \left(DN - \frac{CN}{2} \right) c_{i+1}^k \\
 & d_i c_{i-1}^{k+1} + e_i c_i^{k+1} + f_i c_{i+1}^{k+1} = b_i
 \end{aligned}$$

where

$$CN = \frac{v\Delta t}{\Delta x} = \text{convection number}$$

$$DN = \frac{D\Delta t}{\Delta x^2} = \text{diffusion number}$$

Example problem (contd.)

$$\begin{pmatrix} e & f & 0 & \cdots & 0 & 0 & 0 \\ d & e & f & 0 & \cdots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & \cdots & d & e & f & \cdots & 0 \\ \vdots & \vdots & 0 & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & d & e & f \\ 0 & 0 & \cdots & 0 & 0 & d & e \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_i \\ \vdots \\ c_{n-1} \\ c_n \end{pmatrix}^{k+1} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_i \\ \vdots \\ b_{n-1} \\ b_n \end{pmatrix}$$

Neumann Boundary Condition (2nd type)

- Prescribe a gradient along a portion of the boundary

$$\mathbf{D}\nabla c = q(x, y, z, t)$$

- Where $q(x, y, z, t)$ is a known function of space and time over that portion of boundary

- Example

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} - v \frac{\partial c}{\partial x}$$

$$c(x, 0) = 0, \quad x > 0$$

$$c(0, t) = C_0, \quad t \geq 0$$

$$D \left. \frac{\partial c}{\partial x} \right|_{x=L} = q, \quad t \geq 0$$



Neumann Boundary Condition - FTCS

- Implementation is usually based on central finite differences.
- For example, for FTCS scheme, if the Neuman boundary condition $D(\partial c / \partial x) = q$ is specified at node n then,

$$D \frac{c_{n+1}^k - c_{n-1}^k}{2\Delta x} = q$$

$$c_{n+1}^k = c_{n-1}^k + \frac{2q\Delta x}{D}$$

- The finite difference equations for $\varpi = 0$ become,

$$c_1^{k+1} = \left(DN + \frac{CN}{2} \right) c_0 + (1 - 2DN)c_1^k + \left(DN - \frac{CN}{2} \right) c_2^k$$

$$c_i^{k+1} = \left(DN + \frac{CN}{2} \right) c_{i-1}^k + (1 - 2DN)c_i^k + \left(DN - \frac{CN}{2} \right) c_{i+1}^k$$

$$c_n^{k+1} = \left(DN + \frac{CN}{2} \right) c_{n-1}^k + (1 - 2DN)c_n^k + \left(DN - \frac{CN}{2} \right) c_{n+1}^k$$

$$= (2DN)c_{n-1}^k + (1 - 2DN)c_n^k + \left(DN - \frac{CN}{2} \right) \frac{2q\Delta x}{D}$$

Neumann Boundary Condition – General FDA

- For general FD scheme, the gradient at the boundary node n is given by

$$D \left(\omega \frac{c_{n+1}^{k+1} - c_{n-1}^{k+1}}{2\Delta x} + (1-\omega) \frac{c_{n+1}^k - c_{n-1}^k}{2\Delta x} \right) = q$$

$$c_{n+1}^{k+1} = \frac{\omega c_{n-1}^{k+1} + 2q\Delta x / D - (1-\omega)(c_{n+1}^k - c_{n-1}^k)}{\omega}$$

$$\begin{pmatrix} e_1 & f_1 & & & 0 \\ \ddots & \ddots & \ddots & & \\ & d_i & e_i & f_i & \\ & & \ddots & \ddots & \ddots \\ 0 & & & d_n & e_n \end{pmatrix}$$

- FDA for node n is given by

$$-\omega \left(DN + \frac{CN}{2} \right) c_{n-1}^{k+1} + [1 + 2DN\omega] c_n^{k+1} - \omega \left(DN - \frac{CN}{2} \right) c_{n+1}^{k+1} =$$

$$(1-\omega) \left(DN + \frac{CN}{2} \right) c_{n-1}^k + [1 - 2DN(1-\omega)] c_n^k + (1-\omega) \left(DN - \frac{CN}{2} \right) c_{n+1}^k$$

$$d_n c_{n-1}^{k+1} + e_n c_n^{k+1} + f_n c_{n+1}^{k+1} = b_n$$

Neumann Boundary Condition (contd.)

substituting for c_{n+1}^{k+1} we get,

$$-\omega \left(DN + \frac{CN}{2} \right) c_{n-1}^{k+1} + [1 + 2DN\omega] c_n^{k+1} - \omega \left(DN - \frac{CN}{2} \right) \frac{\omega c_{n-1}^{k+1} + 2q\Delta x / D - (1 - \omega)(c_{n+1}^k - c_{n-1}^k)}{\omega} =$$

$$(1 - \omega) \left(DN + \frac{CN}{2} \right) c_{n-1}^k + [1 - 2DN(1 - \omega)] c_n^k + (1 - \omega) \left(DN - \frac{CN}{2} \right) c_{n+1}^k$$

Grouping terms we get,

$$d_n = -2DN\omega$$

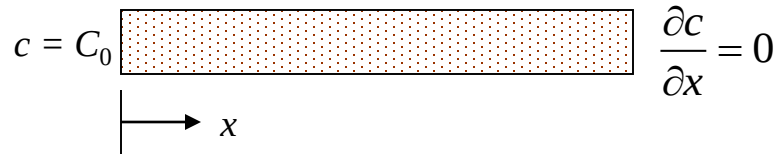
$$e_n = 1 + 2DN\omega$$

$$b_n = 2(1 - \omega)DNc_{n-1}^k + [1 - 2DN(1 - \omega)]c_n^k + \left(DN - \frac{CN}{2} \right) \frac{2q\Delta x}{D}$$

$$\begin{pmatrix} e_1 & f_1 & & & 0 \\ \ddots & \ddots & \ddots & & \\ & d_i & e_i & f_i & \\ & & \ddots & \ddots & \ddots \\ 0 & & & d_n & e_n \end{pmatrix}$$

Dirichlet (left) + Zero Neumann (right)

- Dirichlet at left end ($c = C_0$) and Neumann with $\partial c / \partial x = 0$ at right end.



$c = C_0$

 $\frac{\partial c}{\partial x} = 0$

$\xrightarrow{x}$

$$\begin{pmatrix} e_1 & f_1 & & & 0 \\ \ddots & \ddots & \ddots & & \\ & d_i & e_i & f_i & \\ & & \ddots & \ddots & \ddots \\ 0 & & & d_n & e_n \end{pmatrix}$$

$$d_i = -\omega \left(DN + \frac{CN}{2} \right), \quad e_i = 1 + 2DN\omega, \quad f_i = -\omega \left(DN - \frac{CN}{2} \right), \quad i = 1, \dots, n$$

$$b_i = (1 - \omega) \left(DN + \frac{CN}{2} \right) c_{i-1}^k + [1 - 2DN(1 - \omega)] c_i^k + (1 - \omega) \left(DN - \frac{CN}{2} \right) c_{i+1}^k, \quad i = 2, \dots, n-1$$

$$b_1 = \left(DN + \frac{CN}{2} \right) C_0 + [1 - 2DN(1 - \omega)] c_1^k + (1 - \omega) \left(DN - \frac{CN}{2} \right) c_2^k$$

$$b_n = 2(1 - \omega) DN c_{n-1}^k + [1 - 2DN(1 - \omega)] c_n^k$$

$$d_n = -2DN\omega, \quad e_n = 1 + 2DN\omega$$

Dirichlet (left) + General Neumann (right)

- For general FD scheme

$$D \left(\omega \frac{c_{n+1}^{k+1} - c_{n-1}^{k+1}}{2\Delta x} + (1-\omega) \frac{c_{n+1}^k - c_{n-1}^k}{2\Delta x} \right) = q$$
$$c_{n+1}^{k+1} = \frac{c_{n-1}^{k+1} + 2q\Delta x / D - (1-\omega)(c_{n+1}^k - c_{n-1}^k)}{\omega}$$

$$d_n = -CN$$

$$b_i = (1-\omega) \left(DN + \frac{CN}{2} \right) c_{i-1}^k + [1 - 2DN(1-\omega)] c_i^k + (1-\omega) \left(DN - \frac{CN}{2} \right) c_{i+1}^k, \quad i = 2, \dots, n-1$$

$$b_1 = (1-\omega) \left(DN + \frac{CN}{2} \right) c_0^k + [1 - 2DN(1-\omega)] c_1^k + (1-\omega) \left(DN - \frac{CN}{2} \right) c_2^k$$

$$b_n = (1-\omega) \left(DN + \frac{CN}{2} + \frac{1}{\omega} \right) c_{n-1}^k + [1 - 2DN(1-\omega)] c_n^k + \frac{2q\Delta x}{D}$$

Cauchy (Mixed) Boundary Condition (3rd type)

- Prescribe a concentration and gradient along a portion of the boundary (Special Case: Dankwertz b.c when $q = 0$ or $q = vC_0$)

$$\mathbf{D}\nabla c - \mathbf{v}c = q(x, y, z, t)$$

- Where $q(x, y, z, t)$ is a known function of space and time over that portion of boundary

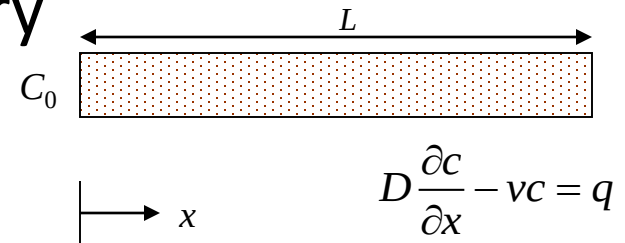
- Example

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} - v \frac{\partial c}{\partial x}$$

$$c(x, 0) = 0, \quad x > 0$$

$$c(0, t) = C_0, \quad t \geq 0$$

$$D \frac{\partial c}{\partial x} - vc \Big|_{x=L} = q, \quad t \geq 0$$



3rd type (right end) - FTCS

- Implementation can be based on central finite differences. For example, for FTCS scheme

$$D \frac{c_{n+1}^k - c_{n-1}^k}{2\Delta x} - v c_n^k = q$$

$$c_{n+1}^k = c_{n-1}^k + \frac{2v c_n^k \Delta x}{D} + \frac{2q \Delta x}{D}$$

- The FDEs become (1st type left end + 3rd type right end),

$$c_1^{k+1} = \left(DN - \frac{CN}{2} \right) c_0 + (1 - 2DN) c_1^k - \left(DN + \frac{CN}{2} \right) c_2^k$$

$$c_i^{k+1} = \left(DN - \frac{CN}{2} \right) c_{i-1}^k + (1 - 2DN) c_i^k - \left(DN + \frac{CN}{2} \right) c_{i+1}^k$$

$$c_n^{k+1} = \left(DN - \frac{CN}{2} \right) c_{n-1}^k + (1 - 2DN) c_n^k - \left(DN + \frac{CN}{2} \right) c_{n+1}^k$$

$$= (-CN) c_{n-1}^k + \left(1 - 2DN + \frac{2v\Delta x}{D} \right) c_n^k - \left(DN + \frac{CN}{2} \right) \frac{2q\Delta x}{D}$$

Mixed Boundary Condition – General FDA

- For general FD scheme, the gradient at the boundary node n is given by

$$D \left(\omega \frac{c_{n+1}^{k+1} - c_{n-1}^{k+1}}{2\Delta x} + (1-\omega) \frac{c_{n+1}^k - c_{n-1}^k}{2\Delta x} \right) - v \left(\omega c_n^{k+1} + (1-\omega) c_n^k \right) = q$$

$$c_{n+1}^{k+1} = \frac{1}{\omega} \left\{ \omega c_{n-1}^{k+1} - (1-\omega) (c_{n+1}^k - c_{n-1}^k) + \frac{2\Delta x}{D} \left[q + v \omega c_n^{k+1} + v(1-\omega) c_n^k \right] \right\}$$

$$\begin{pmatrix} e_1 & f_1 & & & 0 \\ \ddots & \ddots & \ddots & & \\ & d_i & e_i & f_i & \\ & & \ddots & \ddots & \ddots \\ 0 & & & d_n & e_n \end{pmatrix}$$

- FDA for node n is given by

$$-\omega \left(DN + \frac{CN}{2} \right) c_{n-1}^{k+1} + [1 + 2DN\omega] c_n^{k+1} - \omega \left(DN - \frac{CN}{2} \right) c_{n+1}^{k+1} =$$

$$(1-\omega) \left(DN + \frac{CN}{2} \right) c_{n-1}^k + [1 - 2DN(1-\omega)] c_n^k + (1-\omega) \left(DN - \frac{CN}{2} \right) c_{n+1}^k$$

$$d_n c_{n-1}^{k+1} + e_n c_n^{k+1} = b_n$$

Mixed Boundary Condition – General FDA

substituting for c_{n+1}^{k+1} we get,

$$\begin{aligned} & -\omega \left(DN + \frac{CN}{2} \right) c_{n-1}^{k+1} + [1 + 2DN\omega] c_n^{k+1} \\ & - \left(DN - \frac{CN}{2} \right) \left\{ \omega c_{n-1}^{k+1} - (1-\omega)(c_{n+1}^k - c_{n-1}^k) + \frac{2\Delta x}{D} [q + v\omega c_n^{k+1} + v(1-\omega)c_n^k] \right\} \\ & = (1-\omega) \left(DN + \frac{CN}{2} \right) c_{n-1}^k + [1 - 2DN(1-\omega)] c_n^k + (1-\omega) \left(DN - \frac{CN}{2} \right) c_{n+1}^k \end{aligned}$$

Matlab Implementation of 1D CDE

- Governing Equation and initial and boundary conditions – 1D column

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} - v \frac{\partial c}{\partial x}$$

$$c(x, 0) = 0, \quad x > 0$$

$$c(0, t) = C_0, \quad t \geq 0$$

$$c(\infty, t) = 0, \quad t \geq 0$$



$$\text{erfc}(x) = 1 - \frac{1}{\sqrt{2\pi}} \int_0^x e^{-t^2} dt$$

- Analytical Solution

$$\frac{c(x, t)}{C_0} = \frac{1}{2} \left[\text{erfc} \left(\frac{x - vt}{2\sqrt{Dt}} \right) + \exp \left(\frac{vx}{D} \right) \text{erfc} \left(\frac{x + vt}{2\sqrt{Dt}} \right) \right]$$

CN with Dirichlet at both ends

$$\begin{pmatrix} e & f & 0 & \cdots & 0 & 0 & 0 \\ d & e & f & 0 & \cdots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & \cdots & d & e & f & \cdots & 0 \\ \vdots & \vdots & 0 & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & d & e & f \\ 0 & 0 & \cdots & 0 & 0 & d & e \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_i \\ \vdots \\ c_{n-1} \\ c_n \end{pmatrix}^{k+1} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_i \\ \vdots \\ b_{n-1} \\ b_n \end{pmatrix}$$

$$CN = \frac{v\Delta t}{\Delta x}$$

$$DN = \frac{D\Delta t}{\Delta x^2}$$

$$d = -0.5 \left(DN + \frac{CN}{2} \right), \quad e = 1 + DN, \quad f = -0.5 \left(DN - \frac{CN}{2} \right)$$

$$b_i = 0.5 \left(DN + \frac{CN}{2} \right) c_{i-1}^k + [1 - DN] c_i^k + 0.5 \left(DN - \frac{CN}{2} \right) c_{i+1}^k, \quad i = 2, \dots, n-1$$

$$b_1 = 0.5 \left(DN + \frac{CN}{2} \right) C_0 + [1 - DN] c_1^k + 0.5 \left(DN - \frac{CN}{2} \right) c_2^k$$

$$b_n = 0.5 \left(DN + \frac{CN}{2} \right) c_{n-1}^k + [1 - DN] c_n^k$$

1D CDE with Dankwerts BC's

- Governing Equation and initial and boundary conditions – 1D column

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} - v \frac{\partial c}{\partial x}$$

$$vc - D \frac{\partial c}{\partial x} = vC_{in} \quad \boxed{\text{column}} \quad \frac{\partial c}{\partial x} = 0$$



IC : $c = 0$ at $t = 0$ for all x

BC1: $vc - D \frac{\partial c}{\partial x} = vC_{in}$ at $x = 0$ for all t

BC2: $\frac{\partial c}{\partial x} = 0$ at $x = L$ for all t

Crank Nicolson

$$\begin{pmatrix} e & f & 0 & \cdots & 0 & 0 & 0 \\ d & e & f & 0 & \cdots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & \cdots & d & e & f & \cdots & 0 \\ \vdots & \vdots & 0 & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & d & e & f \\ 0 & 0 & \cdots & 0 & 0 & d & e \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_i \\ \vdots \\ c_{n-1} \\ c_n \end{pmatrix}^{k+1} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_i \\ \vdots \\ b_{n-1} \\ b_n \end{pmatrix}$$

$$CN = \frac{v\Delta t}{\Delta x}$$

$$DN = \frac{D\Delta t}{\Delta x^2}$$

$$d_i = -0.5 \left(DN + \frac{CN}{2} \right), \quad e_i = 1 + DN, \quad f_i = -0.5 \left(DN - \frac{CN}{2} \right), \quad i = 1, \dots, n$$

$$e_1 = e_1 - d_1 \frac{2v\Delta x}{D}, \quad f_1 = f_1 + d_1, \quad d_n = d_n + f_n$$

$$c_0^k = c_2^k + \frac{2v\Delta x}{D} (C_{in} - c_1^k), \quad c_{n+1}^k = c_{n-1}^k$$

$$b_i = 0.5 \left(DN + \frac{CN}{2} \right) c_{i-1}^k + [1 - DN] c_i^k + 0.5 \left(DN - \frac{CN}{2} \right) c_{i+1}^k, \quad i = 1, \dots, n$$

$$b_1 = b_1 - d_1 \frac{2v\Delta x C_{in}}{D}$$

Inhomogeneous 1D CDE

- Inhomogeneous 1D CDE

$$\frac{\partial c}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial c}{\partial x} \right) - v \frac{\partial c}{\partial x}$$

- FTCS scheme

$$\frac{c_i^{k+1} - c_i^k}{\Delta t} = \frac{D_{i+1/2} \left(\frac{c_{i+1}^k - c_i^k}{\Delta x} \right) - D_{i-1/2} \left(\frac{c_i^k - c_{i-1}^k}{\Delta x} \right)}{\Delta x} - v_i \left(\frac{c_{i+1}^k - c_{i-1}^k}{2\Delta x} \right)$$

$$c_i^{k+1} = c_i^k + \frac{\Delta t}{2\Delta x^2} \left[(D_{i+1} + D_i) c_{i+1}^k - (D_{i+1} + 2D_i + D_i) c_i^k + (D_i + D_{i-1}) c_{i-1}^k \right] - \frac{v_i \Delta t}{2\Delta x} (c_{i+1}^k - c_{i-1}^k)$$

2D Convection Diffusion Equation

- 2D Homogeneous CDE

$$\frac{\partial c}{\partial t} = \mathbf{D} \nabla^2 c - \mathbf{v} \nabla c$$

$$\frac{\partial c}{\partial t} = \left(D_{xx} \frac{\partial^2 c}{\partial x^2} + D_{xy} \frac{\partial^2 c}{\partial x \partial y} + D_{yy} \frac{\partial^2 c}{\partial y^2} + D_{yx} \frac{\partial^2 c}{\partial y \partial x} \right) - \left(v_x \frac{\partial c}{\partial x} + v_y \frac{\partial c}{\partial y} \right)$$

$$D_{xx} = D_m + \alpha_L \frac{v_x^2}{V} + \alpha_T \frac{v_y^2}{V}, \quad D_{yy} = D_m + \alpha_T \frac{v_x^2}{V} + \alpha_L \frac{v_y^2}{V}$$

$$D_{xy} = D_{yx} = (\alpha_T - \alpha_L) \frac{v_x v_y}{V}, \quad V = \sqrt{v_x^2 + v_y^2}$$

2D Convection Diffusion Equation

- When the longitudinal mean flow direction is aligned with the x -axis (common situation), the off-diagonal components of the dispersion tensor are negligible

$$\frac{\partial c}{\partial t} = \left(D_{xx} \frac{\partial^2 c}{\partial x^2} + D_{yy} \frac{\partial^2 c}{\partial y^2} \right) - \left(v_x \frac{\partial c}{\partial x} + v_y \frac{\partial c}{\partial y} \right)$$

$$D_{xx} = D_m + \alpha_L V, \quad D_{yy} = D_m + \alpha_T V$$

$$V = \sqrt{v_x^2 + v_y^2}$$

Implementation of general FDE for 2D CDE

$$\begin{aligned} \frac{c_{i,j}^{k+1} - c_{i,j}^k}{\Delta t} = & D_{xx} \left(\omega \frac{c_{i+1,j}^{k+1} - 2c_{i,j}^{k+1} + c_{i-1,j}^{k+1}}{\Delta x^2} + (1-\omega) \frac{c_{i+1,j}^k - 2c_{i,j}^k + c_{i-1,j}^k}{\Delta x^2} \right) \\ & - v_x \left(\omega \frac{c_{i+1,j}^{k+1} - c_{i-1,j}^{k+1}}{2\Delta x} + (1-\omega) \frac{c_{i+1,j}^k - c_{i-1,j}^k}{2\Delta x} \right) + \\ & D_{yy} \left(\omega \frac{c_{i,j+1}^{k+1} - 2c_{i,j}^{k+1} + c_{i,j-1}^{k+1}}{\Delta y^2} + (1-\omega) \frac{c_{i,j+1}^k - 2c_{i,j}^k + c_{i,j-1}^k}{\Delta y^2} \right) \\ & - v_y \left(\omega \frac{c_{i,j+1}^{k+1} - c_{i,j-1}^{k+1}}{2\Delta y} + (1-\omega) \frac{c_{i,j+1}^k - c_{i,j-1}^k}{2\Delta y} \right) \end{aligned}$$

$\omega = 0$, FTCS

$\omega = 1$, BTCS

$\omega = 0.5$, CN

Implementation of FTCS, BTCS, CN for 2D CDE

- For any node i,j FDA can be rewritten as,

$$\begin{aligned}
 & -\omega \left(DN_x + \frac{CN_x}{2} \right) c_{i-1,j}^{k+1} - \omega \left(DN_y + \frac{CN_y}{2} \right) c_{i,j-1}^{k+1} + \left[1 + 2DN_x \omega + 2DN_y \omega \right] c_{i,j}^{k+1} \\
 & -\omega \left(DN_y - \frac{CN_y}{2} \right) c_{i,j+1}^{k+1} - \omega \left(DN_x - \frac{CN_x}{2} \right) c_{i+1,j}^{k+1} = \\
 & (1-\omega) \left(DN_x + \frac{CN_x}{2} \right) c_{i-1,j}^k + (1-\omega) \left(DN_y + \frac{CN_y}{2} \right) c_{i,j-1}^k + \left[1 - 2DN_x (1-\omega) - 2DN_y (1-\omega) \right] c_{i,j}^k \\
 & + (1-\omega) \left(DN_y - \frac{CN_y}{2} \right) c_{i,j+1}^k + (1-\omega) \left(DN_x - \frac{CN_x}{2} \right) c_{i+1,j}^k \\
 & d_{i,j} c_{i-1,j}^{k+1} + e_{i,j} c_{i,j-1}^{k+1} + f_{i,j} c_{i,j}^{k+1} + g_{i,j} c_{i,j+1}^{k+1} + h_{i,j} c_{i+1,j}^{k+1} = b_{i,j}
 \end{aligned}$$

$$\begin{pmatrix}
 \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots \\
 \ddots & \ddots & \ddots & \ddots & \ddots & 0 & \ddots \\
 \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots \\
 d_i & 0 & e_i & f_i & g_i & 0 & h_i \\
 \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots \\
 \ddots & 0 & \ddots & \ddots & \ddots & \ddots & \ddots \\
 \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots
 \end{pmatrix}$$

Implementation of FTCS, BTCS, CN for 2D CDE

- Where

$$d_{i,j} = -\omega \left(DN_x + \frac{CN_x}{2} \right), \quad e_{i,j} = -\omega \left(DN_y + \frac{CN_y}{2} \right), \quad f_{i,j} = 1 + 2DN_x\omega + 2DN_y\omega,$$

$$g_{i,j} = -\omega \left(DN_y - \frac{CN_y}{2} \right), \quad h_{i,j} = -\omega \left(DN_x - \frac{CN_x}{2} \right)$$

$$b_{i,j} = (1-\omega) \left(DN_x + \frac{CN_x}{2} \right) c_{i-1,j}^k + (1-\omega) \left(DN_y + \frac{CN_y}{2} \right) c_{i,j-1}^k + [1 - 2DN_x(1-\omega) - 2DN_y(1-\omega)] c_{i,j}^k \\ + (1-\omega) \left(DN_y - \frac{CN_y}{2} \right) c_{i,j+1}^k + (1-\omega) \left(DN_x - \frac{CN_x}{2} \right) c_{i+1,j}^k, \quad i = 1, \dots, n$$

- The above values need to be modified for boundary nodes based on the appropriate boundary conditions.

Convection Diffusion Reaction Equation

- CDR is a slight modification of the CDE with one or more reaction terms.

$$\frac{\partial c}{\partial t} = \mathbf{D} \nabla^2 c - \mathbf{v} \nabla c - R$$

- For example, if the only reaction is decay then CDR is

$$\frac{\partial c}{\partial t} = \left(D_{xx} \frac{\partial^2 c}{\partial x^2} + D_{yy} \frac{\partial^2 c}{\partial y^2} \right) - \left(v_x \frac{\partial c}{\partial x} + v_y \frac{\partial c}{\partial y} \right) - \lambda c$$

Inhomogeneous 2D CDE

- Inhomogeneous CDE is given by

$$\frac{\partial c}{\partial t} = \nabla(\mathbf{D}\nabla c) - \mathbf{v}\nabla c$$

$$\frac{\partial c}{\partial t} = \frac{\partial}{\partial x} \left(D_{xx} \frac{\partial c}{\partial x} \right) + \frac{\partial}{\partial y} \left(D_{yy} \frac{\partial c}{\partial y} \right) - \left(v_x \frac{\partial c}{\partial x} + v_y \frac{\partial c}{\partial y} \right)$$

$$D_{xx} = D_m + \alpha_L V, \quad D_{yy} = D_m + \alpha_T V$$

$$V = \sqrt{v_x^2 + v_y^2}, \quad v_x = v_x(x, y, t), \quad v_y = v_y(x, y, t)$$

Inhomogeneous 2D Non-linear CDR

- Inhomogeneous CDR with decay and bioremediation reactions is given by

$$\frac{\partial c}{\partial t} = \nabla (\mathbf{D} \nabla c) - \mathbf{v} \nabla c - \lambda c - \frac{k_1 c}{k_2 + c}$$

$$\frac{\partial c}{\partial t} = \frac{\partial}{\partial x} \left(D_{xx} \frac{\partial c}{\partial x} \right) + \frac{\partial}{\partial y} \left(D_{yy} \frac{\partial c}{\partial y} \right) - \left(v_x \frac{\partial c}{\partial x} + v_y \frac{\partial c}{\partial y} \right) - \lambda c - \frac{k_1 c}{k_2 + c}$$

- The last term (bioremediation) is non-linear

Nonlinear CDR - Explicit Linearization

- The simplest approach to solving a nonlinear PDE such as Nonlinear CDR is to lag the non-linear term by a full time step
- This is called full explicit linearization
- Explicit linearization is easy to implement but may not be accurate if the nonlinearity is severe.

Inhomogeneous Non-linear CDR – Full Explicit Linearization

$$\begin{aligned}
 \frac{c_{i,j}^{k+1} - c_{i,j}^k}{\Delta t} = & \frac{1}{\Delta x} \left[D_{xx(i+1/2,j)} \left(\frac{c_{i+1,j}^{k+1/2} - c_{i,j}^{k+1/2}}{\Delta x} \right) - D_{xx(i-1/2,j)} \left(\frac{c_{i,j}^{k+1/2} - c_{i-1,j}^{k+1/2}}{\Delta x} \right) \right] \\
 & + \frac{1}{\Delta y} \left[D_{yy(i+1/2,j)} \left(\frac{c_{i,j+1}^{k+1/2} - c_{i,j}^{k+1/2}}{\Delta y} \right) - D_{yy(i,j-1/2)} \left(\frac{c_{i,j}^{k+1/2} - c_{i,j-1}^{k+1/2}}{\Delta y} \right) \right] \\
 & - \frac{1}{\Delta x} \left(v_{x(i+1/2,j)} c_{i+1/2,j}^{k+1/2} - v_{x(i-1/2,j)} c_{i-1/2,j}^{k+1/2} \right) \\
 & - \frac{1}{\Delta y} \left(v_{y(i,j+1/2)} c_{i,j+1/2}^{k+1/2} - v_{y(i,j-1/2)} c_{i,j-1/2}^{k+1/2} \right) \\
 & - \lambda c_{i,j}^{k+1/2} - \frac{k_1 c_{i,j}^{k+1/2}}{k_2 + c_{i,j}^k}
 \end{aligned}$$

where $c_{i,j}^{k+1/2} = \frac{c_{i,j}^{k+1} + c_{i,j}^k}{2}$

$c_{i+1/2,j}^{k+1/2} = \frac{c_{i+1,j}^{k+1/2} + c_{i,j}^{k+1/2}}{2}$

$v_{i+1/2,j} = \frac{v_{i+1,j} + v_{i,j}}{2}$

$D_{xx(i+1/2,j)} = \frac{D_{xx(i+1,j)} + D_{xx(i,j)}}{2}$ and so on..

Nonlinear CDR - SIA

- However, a more accurate strategy is to apply iteration within each time step to make sure the solution satisfies the nonlinear FDE – this is called sequential iterative algorithm (SIA)

For $k = 1$, number of time steps

setup FDA: $[A] (c)^{k+1,1} = (b)^1$

linearize FDA by setting $(c)^{k+1,0} = (c)^k$ for nonlinear terms

$m=0$

Do while ($m < \text{maxiter}$ and $\text{error} > \text{eps}$)

 solve $[A] (c)^{k+1,m+1} = (b)^{m+1}$

 update $(c)^{k+1,m+1}$ and $(b)^{m+1}$

$\text{error} = |(c)^{k+1,m+1} - (c)^{k+1,m}|_2$

$m=m+1$

End while (m)

End for (k)

Inhomogeneous Non-linear CDR – SIA

$$\begin{aligned}
 \frac{c_{i,j}^{k+1,m+1} - c_{i,j}^k}{\Delta t} = & \frac{1}{\Delta x} \left[D_{xx(i+1,j)} \left(\frac{c_{i+1,j}^{k+1/2,m+1/2} - c_{i,j}^{k+1/2,m+1/2}}{\Delta x} \right) - D_{xx(i-1,j)} \left(\frac{c_{i,j}^{k+1/2} - c_{i-1,j}^{k+1/2}}{\Delta x} \right) \right] \\
 & \frac{1}{\Delta y} \left[D_{yy(i+1,j)} \left(\frac{c_{i,j+1}^{k+1/2,m+1} - c_{i,j}^{k+1/2,m+1}}{\Delta y} \right) - D_{yy(i,j-1)} \left(\frac{c_{i,j}^{k+1/2,m+1} - c_{i,j-1}^{k+1/2,m+1}}{\Delta y} \right) \right] \\
 & - \frac{1}{\Delta x} \left(v_{x(i,j)} \frac{c_{i+1,j}^{k+1/2,m+1} + c_i^{k+1/2,m+1}}{2} - v_{x(i-1,j)} \frac{c_{i,j}^{k+1/2,m+1} - c_{i-1,j}^{k+1/2,m+1}}{2} \right) \\
 & - \frac{1}{\Delta y} \left(v_{y(i,j)} \frac{c_{i,j+1}^{k+1/2,m+1} + c_i^{k+1/2,m+1}}{2} - v_{y(i,j-1)} \frac{c_{i,j}^{k+1/2,m+1} - c_{i,j-1}^{k+1/2,m+1}}{2} \right) - \lambda c_{i,j}^{k+1/2,m+1} - \frac{k_1 c_{i,j}^{k+1/2,m+1}}{k_2 + c_{i,j}^{k+1/2,m}}
 \end{aligned}$$

where $c_{i,j}^{k+1/2,m+1} = \frac{c_{i,j}^{k+1,m+1} + c_{i,j}^k}{2}$ and $c_{i,j}^{k+1/2,m} = \frac{c_{i,j}^{k+1,m} + c_{i,j}^k}{2}$

Nonlinear CDR – Newton's Method

- Nonlinear CDR can also be solved by the Newton's method.
- Like SLA, Newton's method also requires iterations.
- Newton's method requires computation of a Jacobian matrix (called the Jacobian) in each iteration.
- In some cases, the Jacobian needs to be computed numerically.
 - In this case, the Newton's method is called the secant method.

General Secant Method

- Solve nonlinear system of equations of the form

$$f_i(x) = 0 \quad 1 \leq i \leq n$$

- Compute $n \times n$ Jacobian matrix $J(x)$ given by

$$J_{ij}(x) = \frac{\partial f_i(x)}{\partial x_j} = \frac{f_i(x_j + \delta x_j) - f_i(x_j)}{\delta x_j} \quad 1 \leq i, j \leq n$$

- Perform iteration

$$\Delta x_i^k = x_i^{k+1} - x_i^k$$

$$J(x^k) \Delta x_i^k = -f_i(x^k) \quad 1 \leq i \leq n, \quad k \geq 0$$

$$x_i^{k+1} = x_i^k + \Delta x_i^k$$

Modified Secant Method

- Solve nonlinear system of equations of the form

$$f_i(x) = 0 \quad 1 \leq i \leq n$$

- Compute $n \times n$ Jacobian matrix $J(x)$ given by

$$J_{ij}(x) = \frac{\partial f_i(x)}{\partial x_j} = \frac{f_i(x_j + \varepsilon) - f_i(x_j - \varepsilon)}{2\varepsilon} \quad 1 \leq i, j \leq n$$

- Perform iteration

$$\Delta x_i^k = x_i^{k+1} - x_i^k$$

$$J(x^k) \Delta x_i^k = -f_i(x^k) \quad 1 \leq i \leq n, \quad k \geq 0$$

$$x_i^{k+1} = x_i^k + \Delta x_i^k$$

Method of Characteristics

- A commonly applied technique for solving advection dominated CDE is the method of characteristics (MOC)
- The basic principle behind MOC is that the convective component of the solution propagates along characteristics described by the velocity field.
- Rewrite PDE in terms of material derivative and solve for diffusion equation.

Multi-component CDR

- Multi-component CDR's are also common.

$$\frac{\partial c_1}{\partial t} = \frac{\partial}{\partial x} \left(D_{xx} \frac{\partial c_1}{\partial x} \right) + \frac{\partial}{\partial y} \left(D_{yy} \frac{\partial c_1}{\partial y} \right) - \left(v_x \frac{\partial c_1}{\partial x} + v_y \frac{\partial c_1}{\partial y} \right) - \lambda_1 c_1 - \frac{k_{11} c_1 c_2}{k_{12} + c_1}$$
$$\frac{\partial c_2}{\partial t} = \frac{\partial}{\partial x} \left(D_{xx} \frac{\partial c_2}{\partial x} \right) + \frac{\partial}{\partial y} \left(D_{yy} \frac{\partial c_2}{\partial y} \right) - \left(v_x \frac{\partial c_2}{\partial x} + v_y \frac{\partial c_2}{\partial y} \right) - \lambda_2 c_2 - \frac{k_{21} c_1 c_2}{k_{22} + c_2}$$

- Multi-component CDR's can be solved by lagging the other species concentrations and solving sequentially.
- Again iteration need to be applied to make sure the correct solution is attained

PDE toolbox

- Solves 2-D PDEs:
 - Elliptic (e.g., Steady GW flow)
 - Parabolic (e.g., Unsteady GW flow, GW transport)
 - Hyperbolic (e.g., wave eqn)
 - Eigenvalue PDE
- Structured and unstructured meshes
- Dirichlet and Generalized Neuman BCs
- Graphical interface is `pdetool`
- Various functions available to construct and solve the PDE (`initmesh`, `assemblpde`, `parabolic`, `hyperbolic` etc.)

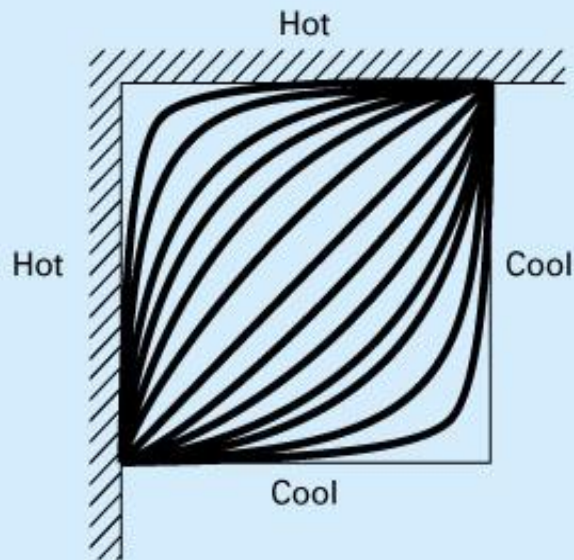
Chapra Book Slides

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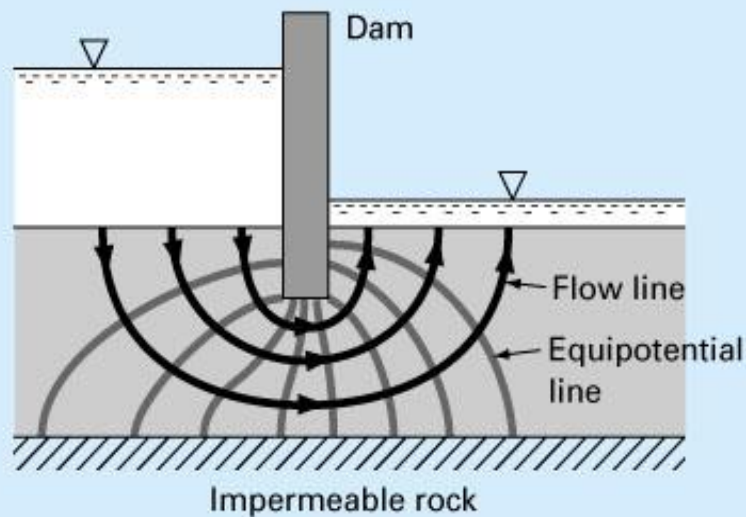
Part 8

Partial Differential Equations

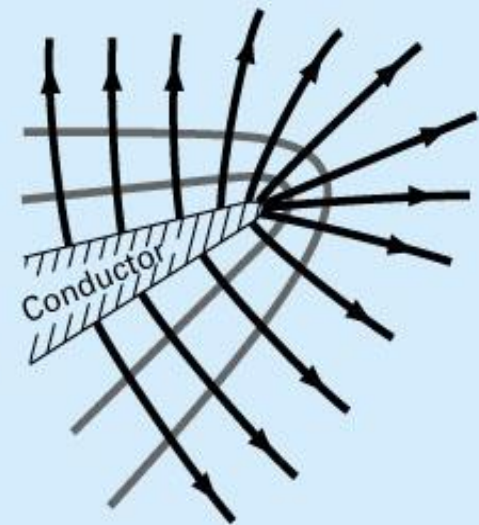
Table PT8.1



(a)

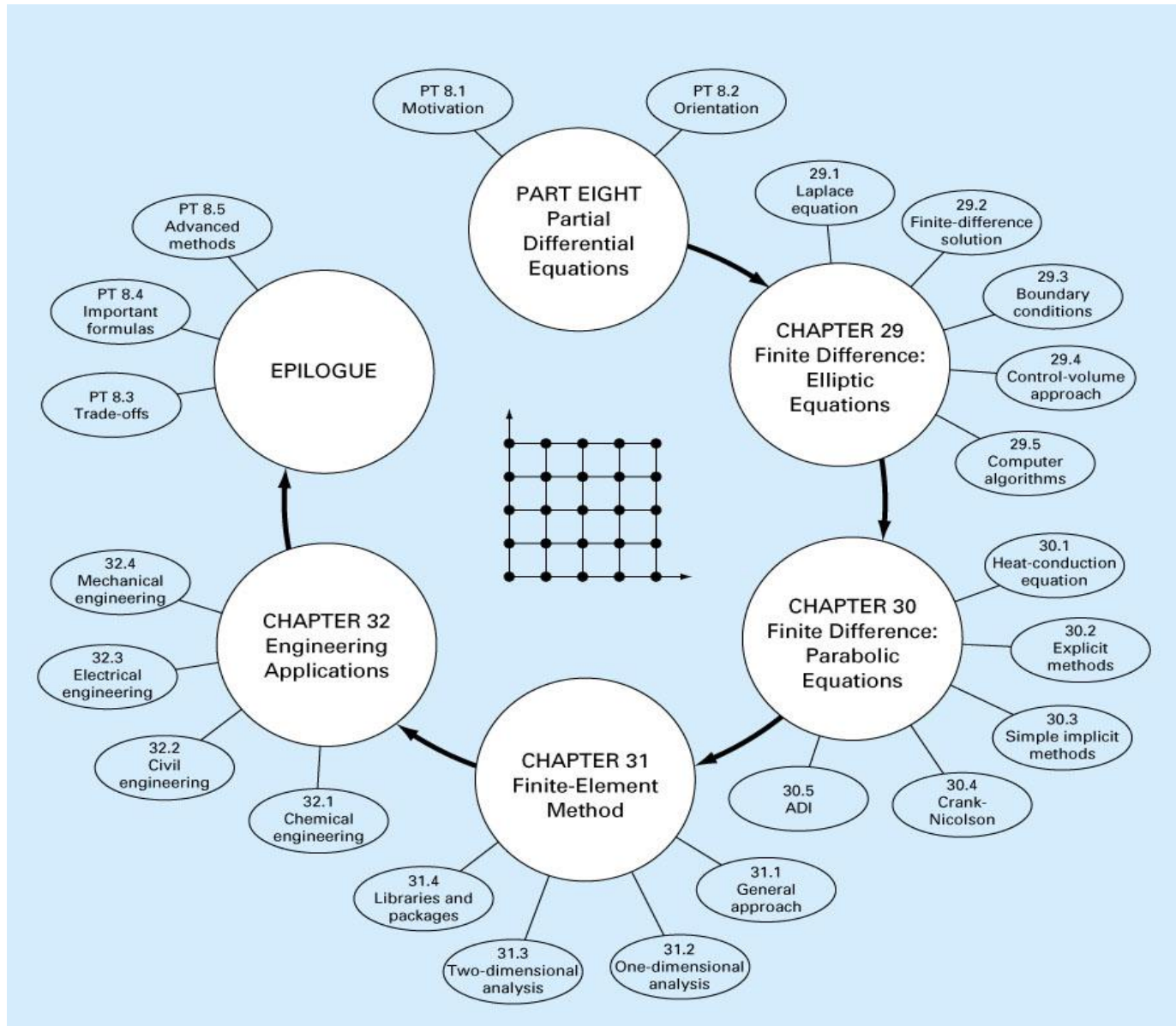


(b)



(c)

Figure PT8.4



Finite Difference: Elliptic Equations

Chapter 29

Solution Technique

- Elliptic equations in engineering are typically used to characterize steady-state, boundary value problems.
- For numerical solution of elliptic PDEs, the PDE is transformed into an algebraic difference equation.
- Because of its simplicity and general relevance to most areas of engineering, we will use a heated plate as an example for solving elliptic PDEs.

Figure 29.1

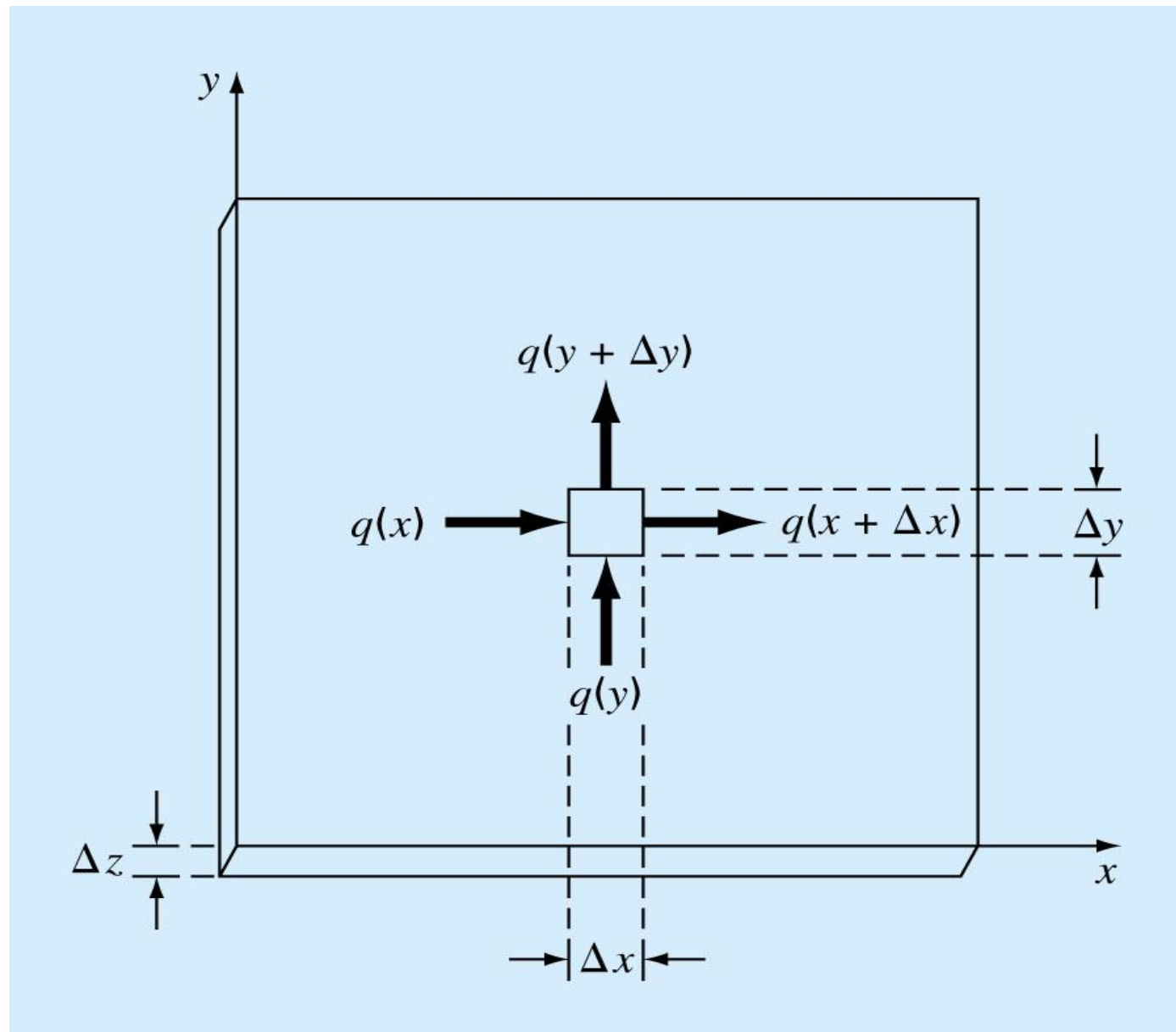
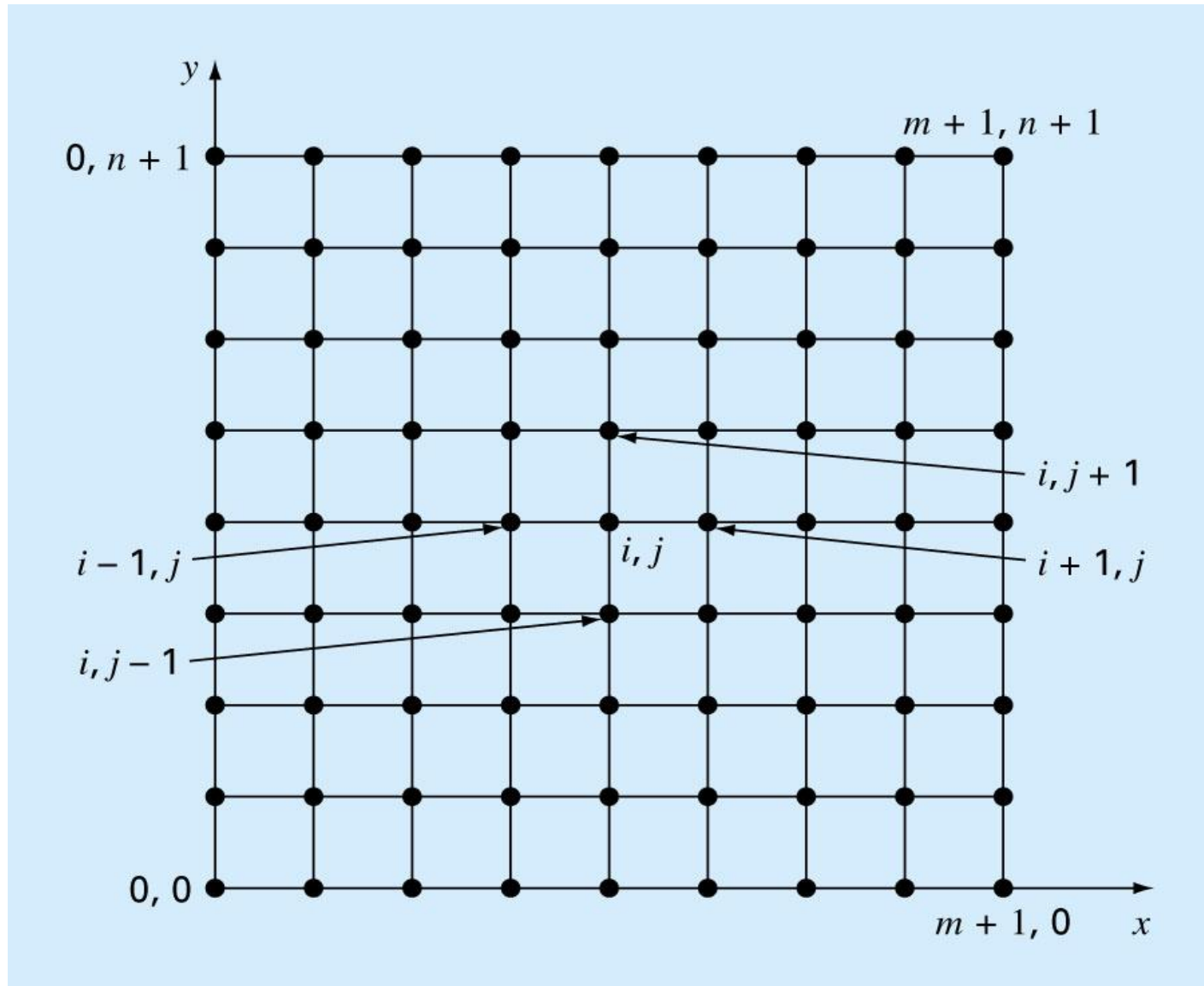


Figure 29.3



The Laplacian Difference Equations/

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

Laplace Equation

$$\frac{\partial^2 T}{\partial x^2} = \frac{T_{i+1,j} - 2T_{i,j} + T_{i-1,j}}{\Delta x^2} \quad O[\Delta(x)^2]$$

$$\frac{\partial^2 T}{\partial y^2} = \frac{T_{i,j+1} - 2T_{i,j} + T_{i,j-1}}{\Delta y^2} \quad O[\Delta(y)^2]$$

$$\frac{T_{i+1,j} - 2T_{i,j} + T_{i-1,j}}{\Delta x^2} + \frac{T_{i,j+1} - 2T_{i,j} + T_{i,j-1}}{\Delta y^2} = 0$$

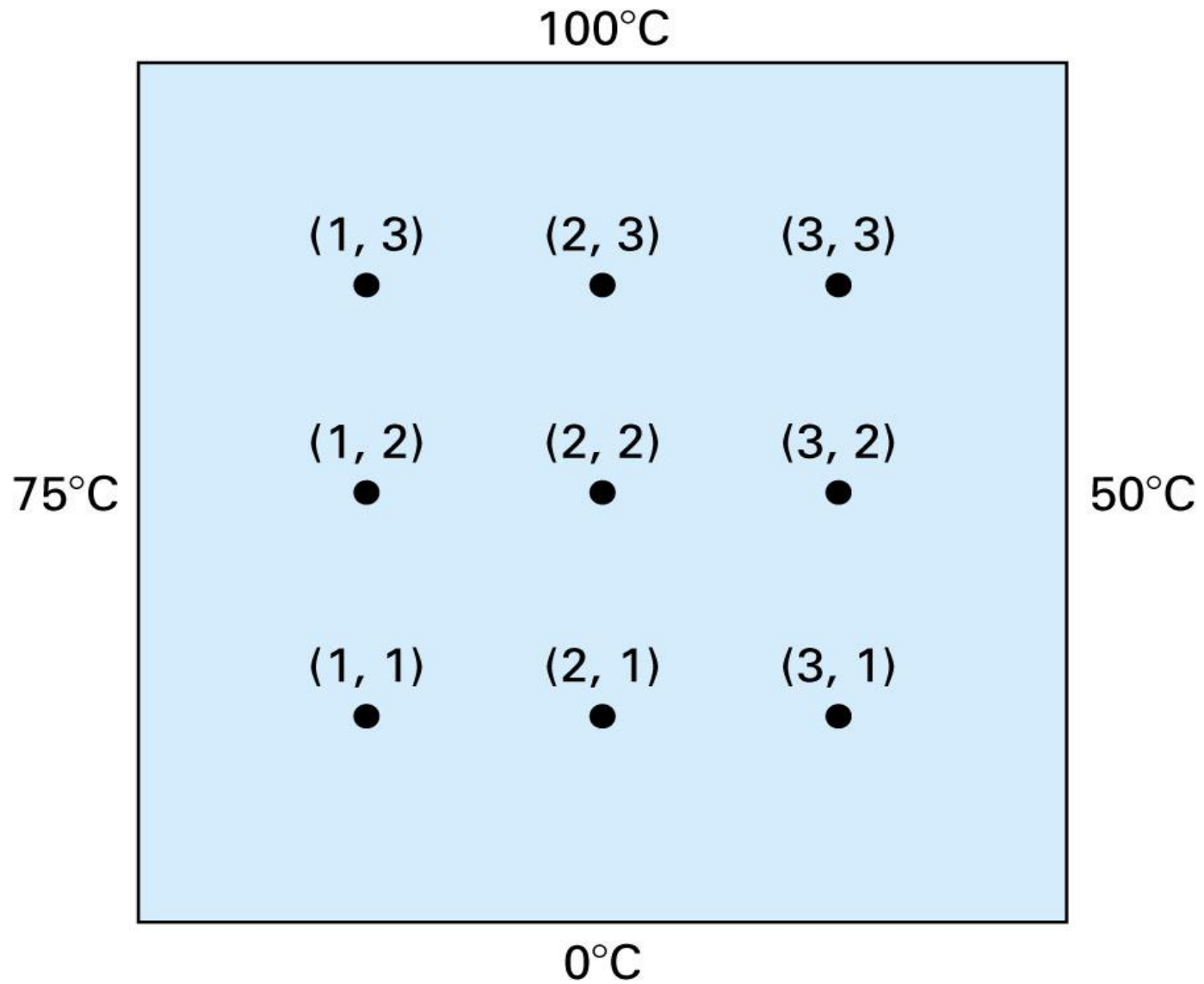
$$\Delta x = \Delta y$$

$$T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1} - 4T_{i,j} = 0$$

Laplacian difference equation.

Holds for all interior points

Figure 29.4



- In addition, boundary conditions along the edges must be specified to obtain a unique solution.
- The simplest case is where the temperature at the boundary is set at a fixed value, *Dirichlet boundary condition*.
- A balance for node (1,1) is:

$$T_{21} + T_{01} + T_{12} + T_{10} - 4T_{11} = 0$$

$$T_{01} = 75$$

$$T_{10} = 0$$

$$-4T_{11} + T_{12} + T_{21} = 0$$
- Similar equations can be developed for other interior points to result a set of simultaneous equations.

- The result is a set of nine simultaneous equations with nine unknowns:

$$\begin{array}{rcl}
 4T_{11} & -T_{21} & -T_{12} & & & & & & = 75 \\
 -T_{11} & +4T_{21} & -T_{13} & & -T_{22} & & & & = 0 \\
 & -T_{21} & +4T_{31} & & & -T_{32} & & & = 50 \\
 -T_{11} & & & +4T_{12} & -T_{22} & & -T_{13} & & = 75 \\
 & -T_{21} & & -T_{12} & +4T_{22} & -T_{32} & & -T_{23} & = 0 \\
 & & -T_{31} & & -T_{22} & +4T_{32} & & -T_{33} & = 50 \\
 & & & -T_{12} & & & +4T_{13} & -T_{23} & = 175 \\
 & & & & -T_{22} & & -T_{13} & +4T_{23} & -T_{33} & = 100 \\
 & & & & & -T_{32} & & -T_{23} & +4T_{33} & = 150
 \end{array}$$

The Liebmann Method/

- Most numerical solutions of Laplace equation involve systems that are very large.
- For larger size grids, a significant number of terms will be zero.
- For such sparse systems, most commonly employed approach is *Gauss-Seidel*, which when applied to PDEs is also referred as *Liebmann's method*.

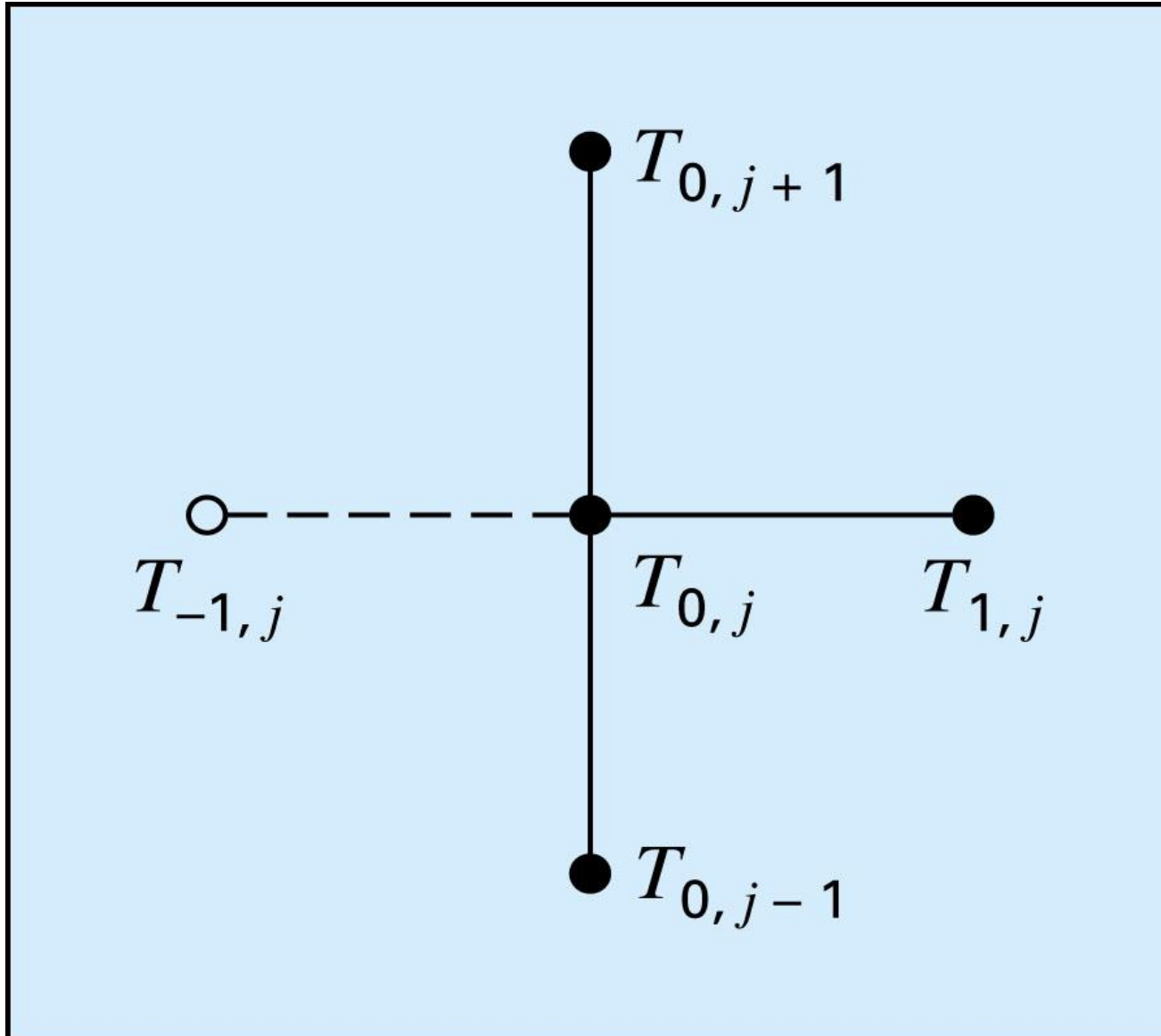
Boundary Conditions

- We will address problems that involve boundaries at which the derivative is specified and boundaries that are irregularly shaped.

Derivative Boundary Conditions/

- Known as a *Neumann boundary condition*.
- For the heated plate problem, heat flux is specified at the boundary, rather than the temperature.
- If the edge is insulated, this derivative becomes zero.

Figure 29.7



$$T_{1,j} + T_{-1,j} + T_{0,j+1} + T_{0,j-1} - 4T_{0,j} = 0$$

$$\frac{\partial T}{\partial x} \cong \frac{T_{1,j} - T_{-1,j}}{2\Delta x}$$

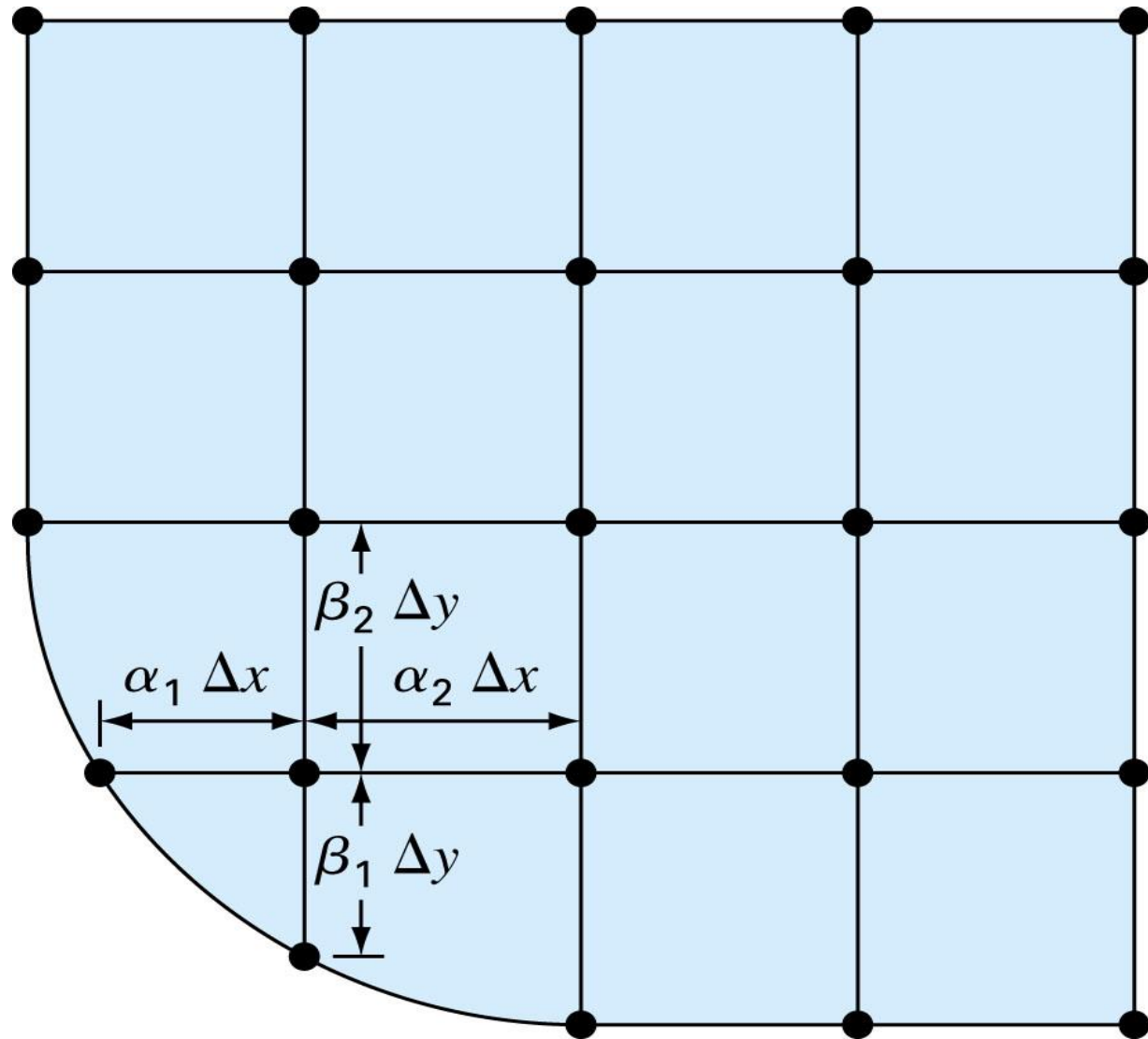
$$T_{-1,j} = T_{1,j} - 2\Delta x \frac{\partial T}{\partial x}$$

$$2T_{1,j} - 2\Delta x \frac{\partial T}{\partial x} + T_{0,j+1} + T_{0,j-1} - 4T_{0,j} = 0$$

- Thus, the derivative has been incorporated into the balance.
- Similar relationships can be developed for derivative boundary conditions at the other edges.

Irregular Boundaries

- Many engineering problems exhibit irregular boundaries.



- First derivatives in the x direction can be approximated as:

$$\left(\frac{\partial T}{\partial x}\right)_{i-1,i} \cong \frac{T_{i,j} - T_{i-1,j}}{\alpha_1 \Delta x}$$

$$\left(\frac{\partial T}{\partial x}\right)_{i,i+1} \cong \frac{T_{i+1,j} - T_{i,j}}{\alpha_2 \Delta x}$$

$$\frac{\partial^2 T}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial T}{\partial x} \right) = \frac{\left(\frac{\partial T}{\partial x}\right)_{i,i+1} - \left(\frac{\partial T}{\partial x}\right)_{i-1,i}}{\frac{\alpha_1 \Delta x + \alpha_2 \Delta x}{2}}$$

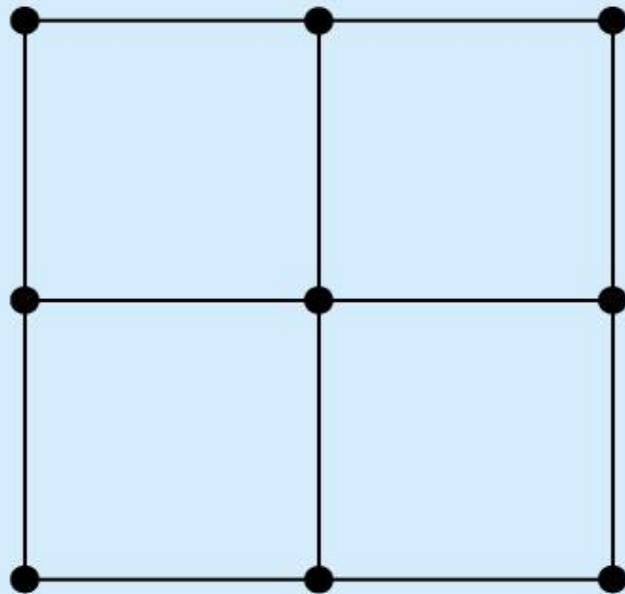
$$\frac{\partial^2 T}{\partial x^2} = 2 \frac{\frac{T_{i,j} - T_{i-1,j}}{\alpha_1 \Delta x} - \frac{T_{i+1,j} - T_{i,j}}{\alpha_2 \Delta x}}{\alpha_1 \Delta x + \alpha_2 \Delta x}$$

$$\frac{\partial^2 T}{\partial x^2} = \frac{2}{\Delta x^2} \left[\frac{T_{i-1,j} - T_{i,j}}{\alpha_1 (\alpha_1 + \alpha_2)} + \frac{T_{i+1,j} - T_{i,j}}{\alpha_2 (\alpha_1 + \alpha_2)} \right]$$

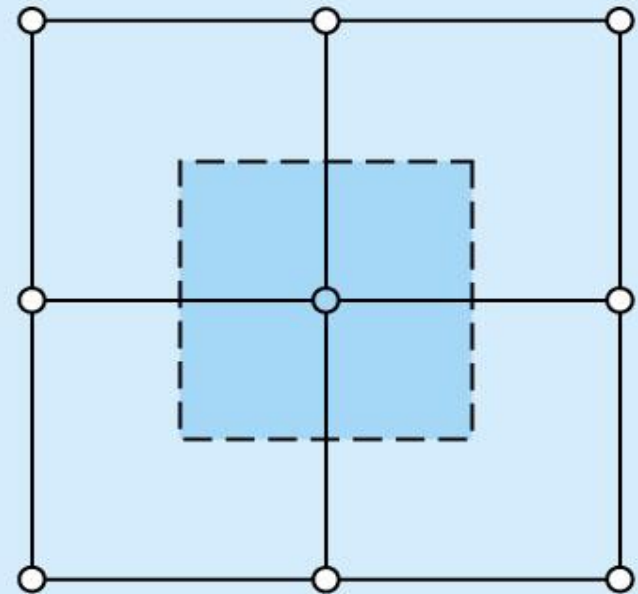
- A similar equation can be developed in the y direction.

Control-Volume Approach

Figure 29.12

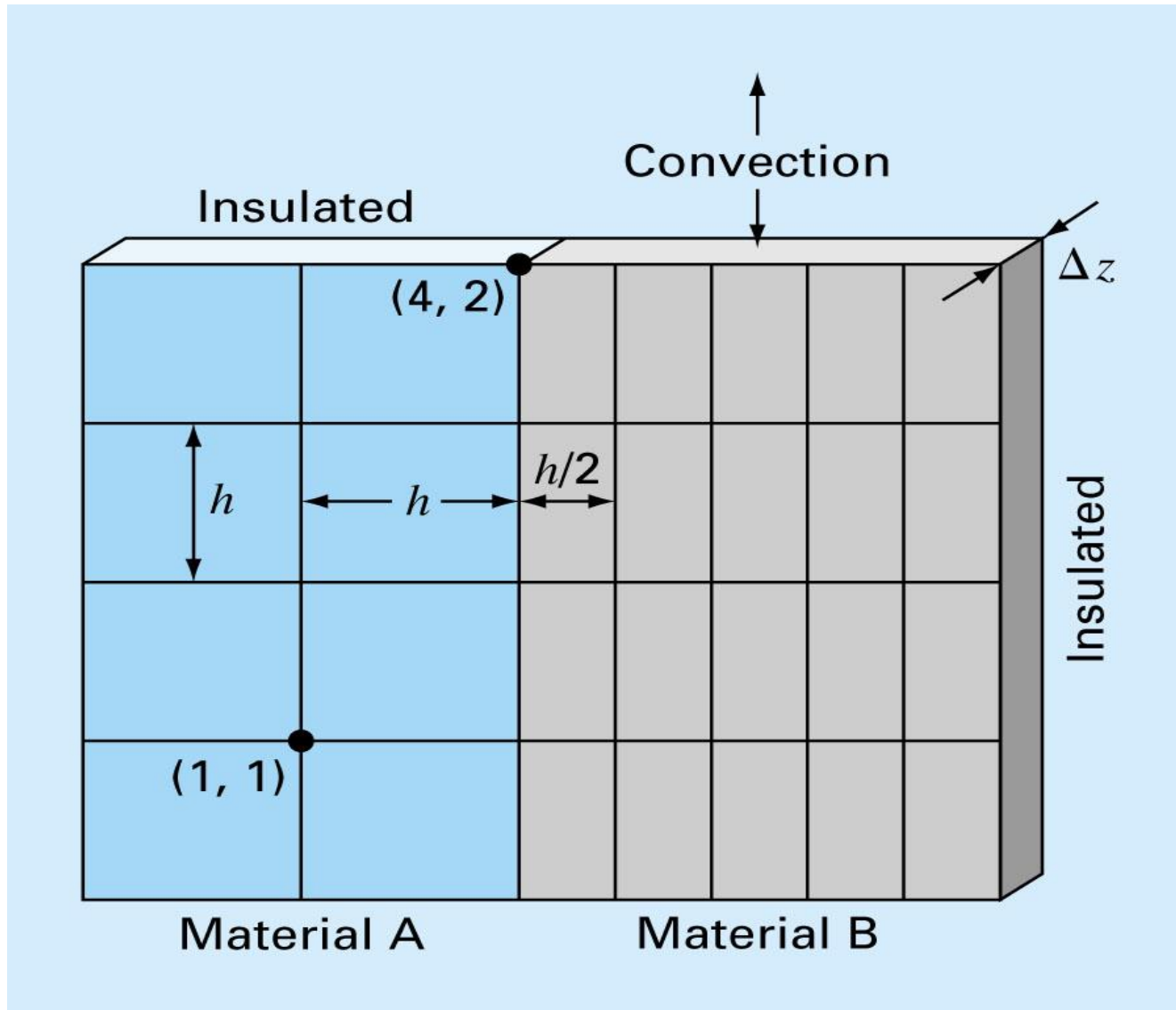


(a) Pointwise, finite-difference approach



(b) Control-volume approach

Figure 29.13



- The *control-volume approach* resembles the point-wise approach in that points are determined across the domain.
- In this case, rather than approximating the PDE at a point, the approximation is applied to a volume surrounding the point.

Finite Difference: Parabolic Equations

Chapter 30

- Parabolic equations are employed to characterize time-variable (*unsteady-state*) problems.
- Conservation of energy can be used to develop an *unsteady-state* energy balance for the differential element in a long, thin insulated rod.

Figure 30.1

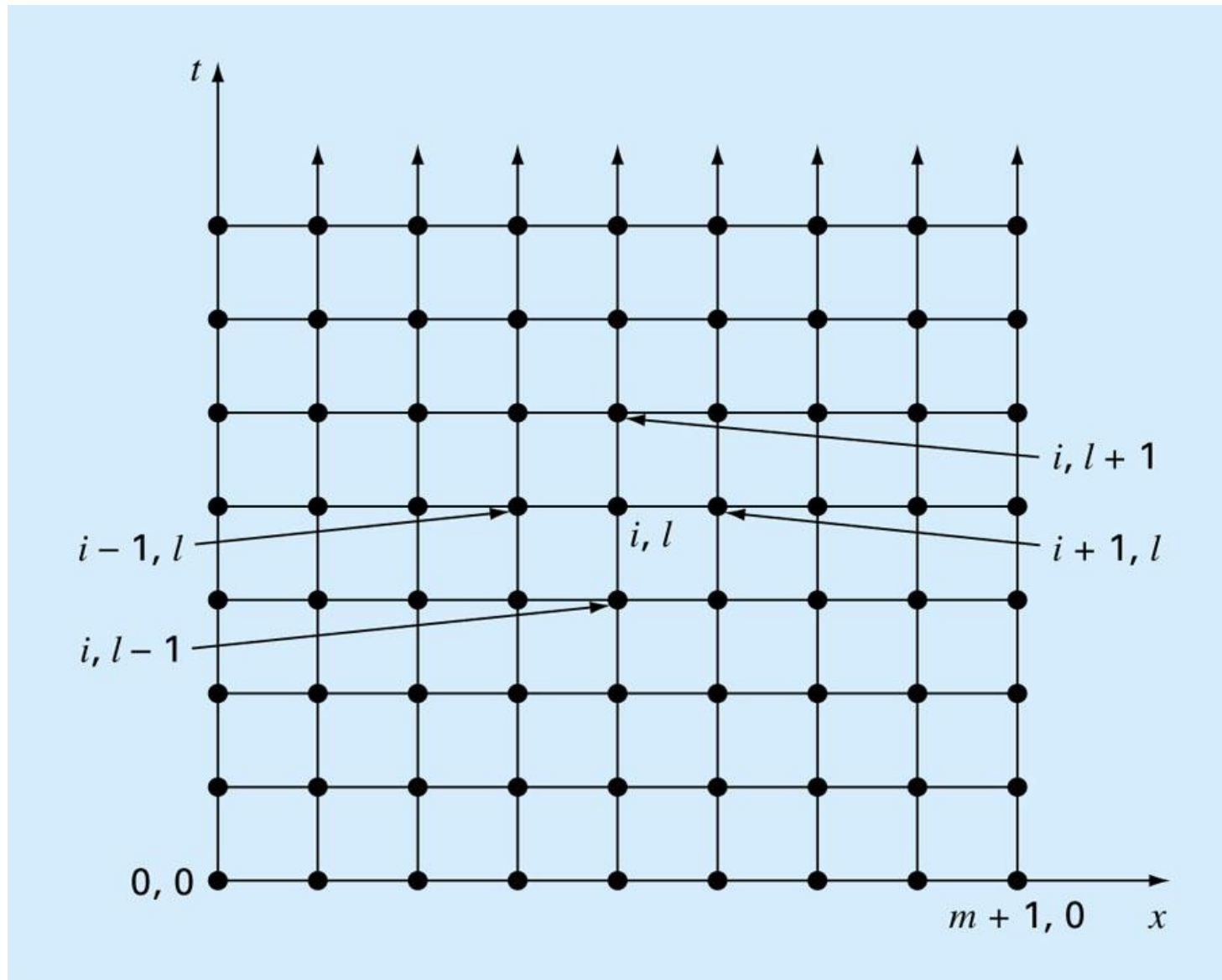


- Energy balance together with Fourier's law of heat conduction yields *heat-conduction equation*:

$$k \frac{\partial^2 T}{\partial x^2} = \frac{\partial T}{\partial t}$$

- Just as elliptic PDEs, parabolic equations can be solved by substituting finite divided differences for the partial derivatives.
- In contrast to elliptic PDEs, we must now consider changes in time as well as in space.
- Parabolic PDEs are temporally open-ended and involve new issues such as stability.

Figure 30.2



Explicit Methods

- The heat conduction equation requires approximations for the second derivative in space and the first derivative in time:

$$\frac{\partial^2 T}{\partial x^2} = \frac{T_{i+1}^l - 2T_i^l + T_{i-1}^l}{\Delta x^2}$$

$$\frac{\partial T}{\partial t} = \frac{T_i^{l+1} - T_i^l}{\Delta t}$$

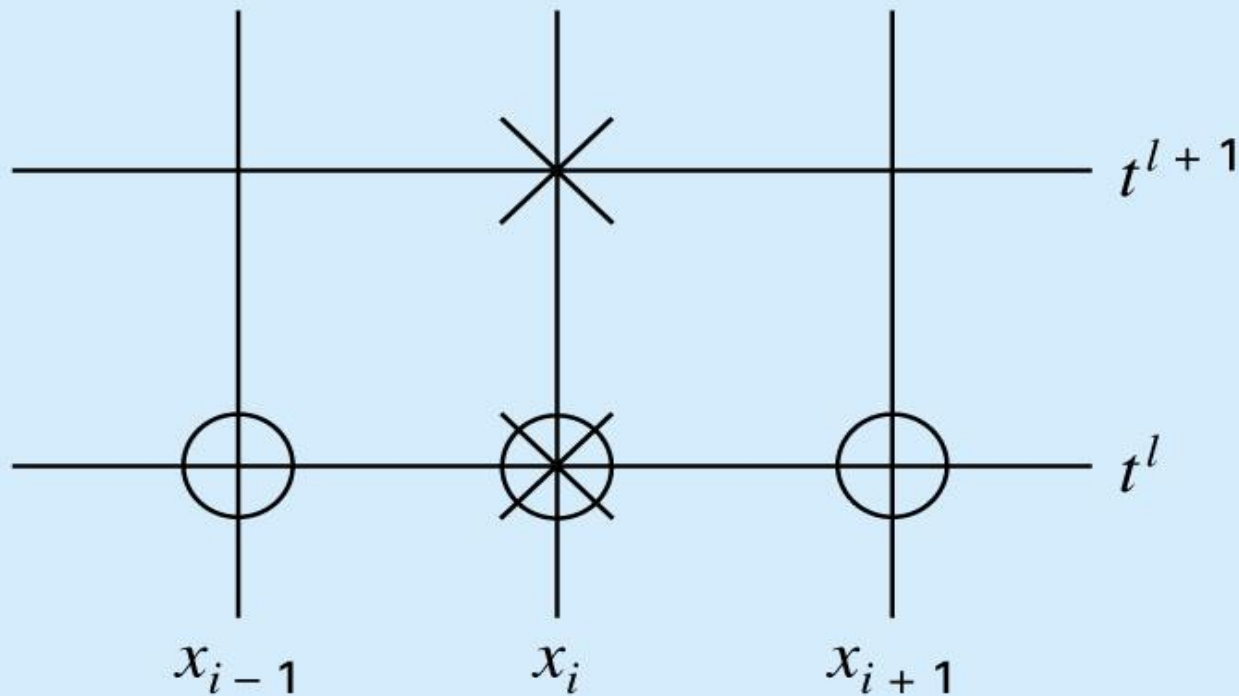
$$k \frac{T_{i+1}^l - 2T_i^l + T_{i-1}^l}{(\Delta x)^2} = \frac{T_i^{l+1} - T_i^l}{\Delta t}$$

$$T_i^{l+1} = T_i^l + \lambda (T_{i+1}^l - 2T_i^l + T_{i-1}^l) \quad \lambda = \frac{k\Delta t}{(\Delta x)^2}$$

- This equation can be written for all interior nodes on the rod.
- It provides an explicit means to compute values at each node for a future time based on the present values at the node and its neighbors.

Figure 30.3

✕ Grid point involved in time difference
○ Grid point involved in space difference



Convergence and Stability/

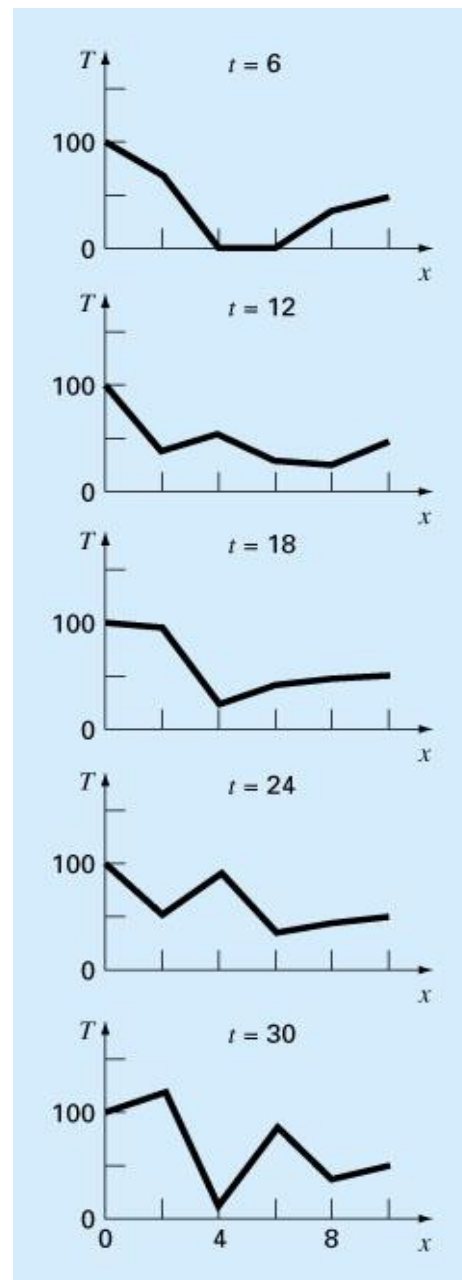
- Convergence means that as Δx and Δt approach zero, the results of the finite difference method approach the true solution.
- Stability means that errors at any stage of the computation are not amplified but are attenuated as the computation progresses.
- The explicit method is both convergent and stable if

$$\lambda \leq 1/2$$

or

$$\Delta t \leq \frac{1}{2} \frac{\Delta x^2}{k}$$

Figure 30.5



Derivative Boundary Conditions/

- As was the case for elliptic PDEs, derivative boundary conditions can be readily incorporated into parabolic equations.

$$T_0^{l+1} = T_0^l + \lambda(T_1^l - 2T_0^l + T_{-1}^l)$$

- Thus an imaginary point is introduced at $i = -1$, providing a vehicle for incorporating the derivative boundary condition into the analysis.

A simple Implicit Method

- Implicit methods overcome difficulties associated with explicit methods at the expense of somewhat more complicated algorithms.
- In implicit methods, the spatial derivative is approximated at an advanced time interval $l+1$:

$$\frac{\partial^2 T}{\partial x^2} \cong \frac{T_{i+1}^{l+1} - 2T_i^{l+1} + T_{i-1}^{l+1}}{(\Delta x)^2}$$

which is second-order accurate.

$$k \frac{T_{i+1}^{l+1} - 2T_i^{l+1} + T_{i-1}^{l+1}}{(\Delta x)^2} = \frac{T_i^{l+1} - T_i^l}{\Delta t}$$

$$- \lambda T_{i-1}^{l+1} + (1 + 2\lambda)T_i^{l+1} - \lambda T_{i+1}^{l+1} = T_i^l$$

This eqn. applies to all but the first and the last interior nodes, which must be modified to reflect the boundary conditions:

$$T_0^{l+1} = f_0(t^{l+1})$$

$$(1 + 2\lambda)T_1^{l+1} - \lambda T_2^{l+1} = T_1^l + f_0(t^{l+1})$$

$$i = m$$

$$(1 + 2\lambda)T_m^{l+1} - \lambda T_{m-1}^{l+1} = T_m^l + f_{m+1}(t^{l+1})$$

Resulting m unknowns and m linear algebraic equations

Figures 30.6

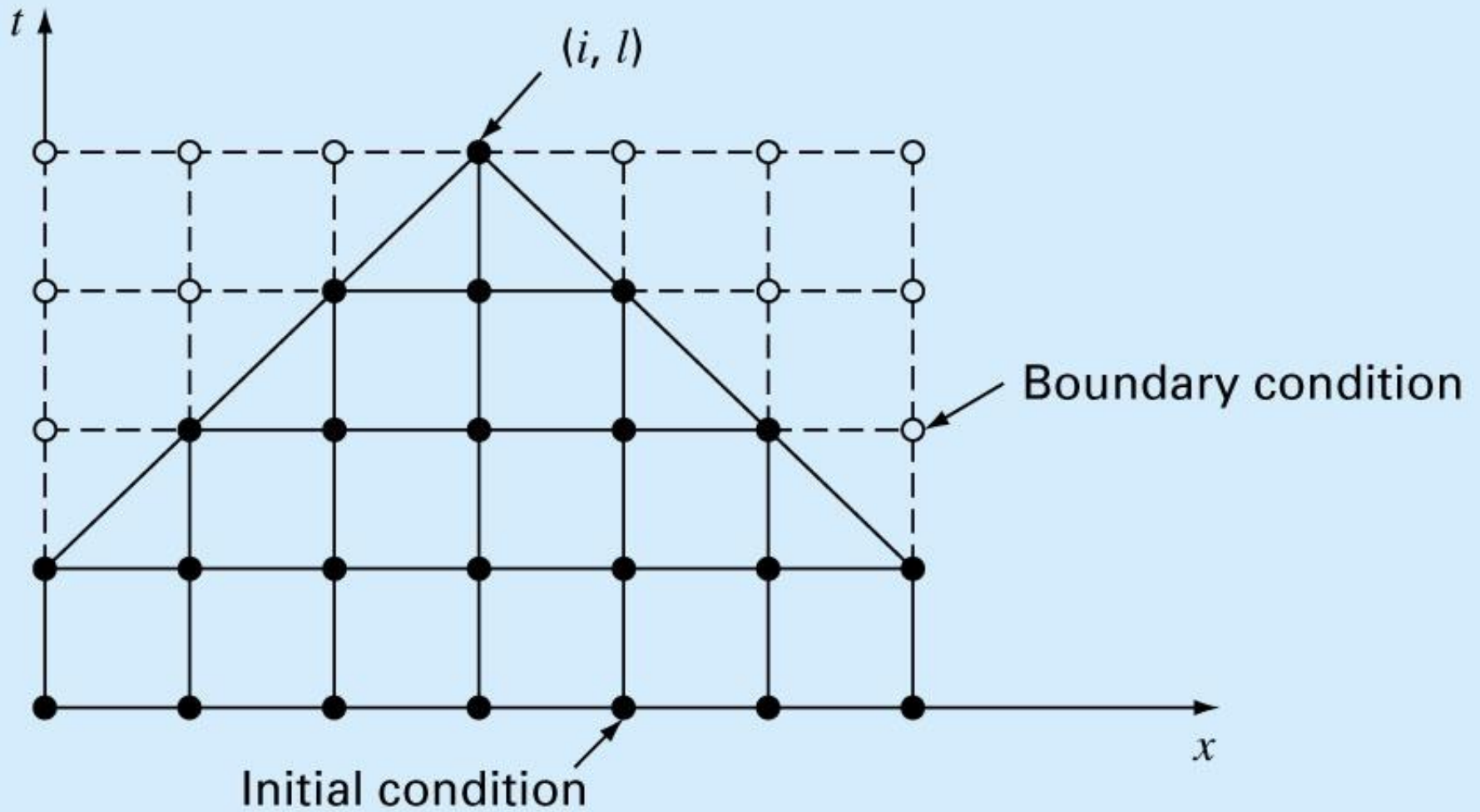
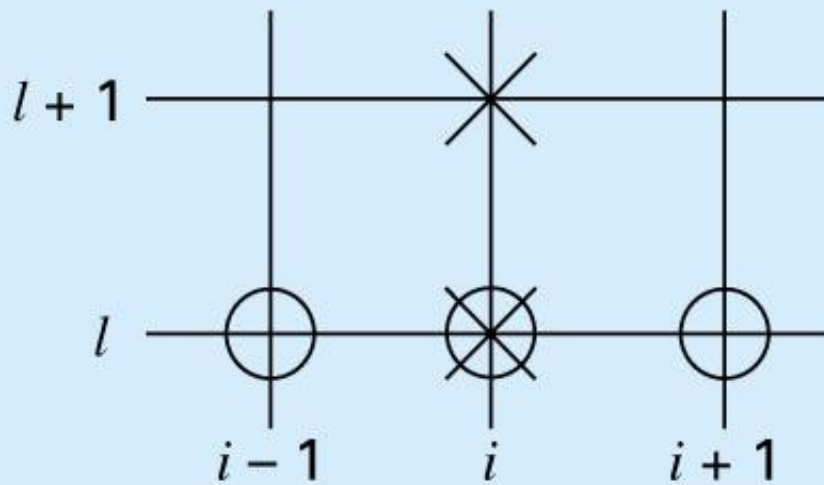
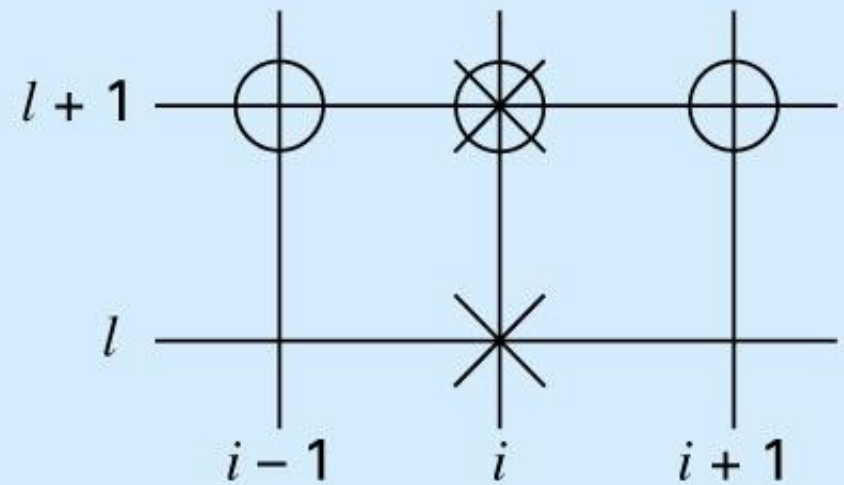


Figure 30.7

 Grid point involved in time difference
 Grid point involved in space difference



(a) Explicit



(b) Implicit

The Crank-Nicolson Method

- Provides an alternative implicit scheme that is second order accurate both in space and time.
- To provide this accuracy, difference approximations are developed at the midpoint of the time increment:

$$\frac{\partial T}{\partial t} \cong \frac{T_i^{l+1} - T_i^l}{\Delta t}$$
$$\frac{\partial^2 T}{\partial x^2} \cong \frac{1}{2} \left[\frac{T_{i+1}^l - 2T_i^l + T_{i-1}^l}{(\Delta x)^2} + \frac{T_{i+1}^{l+1} - 2T_i^{l+1} + T_{i-1}^{l+1}}{(\Delta x)^2} \right]$$

Parabolic Equations in Two Spatial Dimensions

- For two dimensions the heat-conduction equation can be applied more than one spatial dimension:

$$\frac{\partial T}{\partial t} = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)$$

Standard Explicit and Implicit Schemes/

- An *explicit* solution can be obtained by substituting finite-differences approximations for the partial derivatives. However, this approach is limited by a stringent stability criterion, thus increases the required computational effort.

- The direct application of *implicit* methods leads to solution of $m \times n$ simultaneous equations.
- When written for two or three dimensions, these equations lose the valuable property of being tridiagonal and require very large matrix storage and computation time.

The ADI Scheme/

- The alternating-direction implicit, or ADI, scheme provides a means for solving parabolic equations in two spatial dimensions using tridiagonal matrices.
- Each time increment is executed in two steps.
- For the first step, heat conduction equation is approximated by:

$$\frac{T_{i,j}^{l+1/2} - T_{i,j}^l}{\Delta t / 2} = k \left[\frac{T_{i+1,j}^l - 2T_{i,j}^l + T_{i-1,j}^l}{(\Delta x)^2} + \frac{T_{i,j+1}^{l+1/2} - 2T_{i,j}^{l+1/2} + T_{i,j-1}^{l+1/2}}{(\Delta y)^2} \right]$$

- Thus the approximation of partial derivatives are written explicitly, that is at the base point t^l where temperatures are known. Consequently only the three temperature terms in each approximation are unknown.

Figure 30.10

